



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

16-04-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	B,D	B,C,D	B,C,D	A,C,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	8.00	9.00	6.00	2.00	0.75	7.00	6.04

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B	B	A,B,C	B,C	A,B,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	5.00	5.33	10.00	6.00	9.00	5.00	84.40

PART-3 : MATHEMATICS

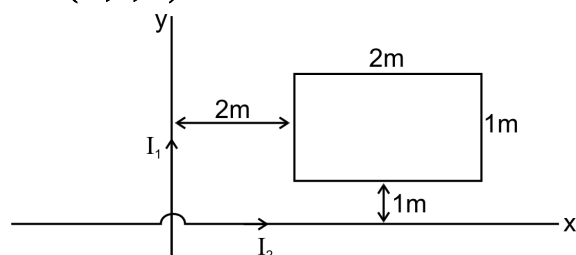
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,B	C,D	A,B,C	A,B,C	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	A	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	3.00	1.00	3.00	3.00	2.00	9.00	1.20

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C,D)



$$I_1 = 2I_0 \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$I_2 = I_0 \cos\left(\omega t + \frac{\pi}{4}\right)$$

$$\phi_1 = \frac{\mu_0 I_1}{2\pi} \int_2^4 \frac{1dx}{x} = \frac{\mu_0 I_1}{2\pi} \ln 2$$

$$\phi_2 = \frac{\mu_0 I_2}{2\pi} \int_1^2 \frac{2dy}{y} = \frac{\mu_0 I_2}{\pi} \ln 2$$

$$\phi_{\text{Total}} = \frac{\mu_0 I_1 \ln 2}{2\pi} - \frac{\mu_0 I_2}{\pi} \ln 2$$

$$= \frac{\mu_0 \ln 2}{\pi} \left[\frac{I_1}{2} - I_2 \right]$$

$$= \frac{\mu_0 \ln 2}{\pi} \left[I_0 \sin\left(\omega t + \frac{\pi}{4}\right) - I_0 \cos\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$= \frac{\mu_0 \ln 2 I_0}{\pi} \left[\sin\left(\omega t + \frac{\pi}{4}\right) - \cos\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 I_0 \omega \ln 2}{\pi} \left[\cos\left(\omega t + \frac{\pi}{4}\right) + \sin\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$\varepsilon_{\text{max.}} = \frac{\sqrt{2} \mu_0 I_0 \omega \ln 2}{\pi}$$

$$Q_{\text{max}} = \frac{\sqrt{2} \mu_0 I_0 \omega \ln 2}{\pi} C$$

$$Q = CE$$

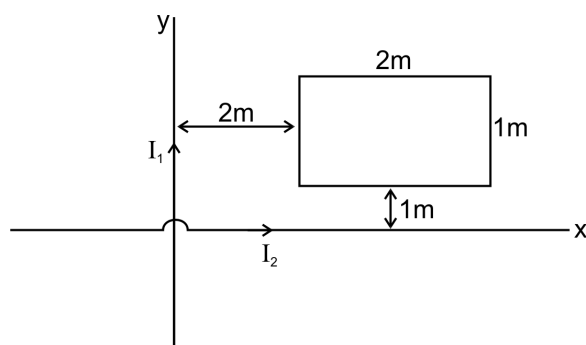
$$\frac{dQ}{dt} = C\varepsilon = C \frac{\mu_0 I_0 \omega^2 \ln 2}{\pi} \left[-\sin\left(\omega t + \frac{\pi}{4}\right) + \cos\left(\omega t + \frac{\pi}{4}\right) \right]$$

Current in frame at $t = \frac{\pi}{\omega}$ is zero.

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C,D)



$$I_1 = 2I_0 \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$I_2 = I_0 \cos\left(\omega t + \frac{\pi}{4}\right)$$

$$\phi_1 = \frac{\mu_0 I_1}{2\pi} \int_2^4 \frac{1dx}{x} = \frac{\mu_0 I_1}{2\pi} \ln 2$$

$$\phi_2 = \frac{\mu_0 I_2}{2\pi} \int_1^2 \frac{2dy}{y} = \frac{\mu_0 I_2}{\pi} \ln 2$$

$$\phi_{\text{Total}} = \frac{\mu_0 I_1 \ln 2}{2\pi} - \frac{\mu_0 I_2}{\pi} \ln 2$$

$$= \frac{\mu_0 \ln 2}{\pi} \left[\frac{I_1}{2} - I_2 \right]$$

$$= \frac{\mu_0 \ln 2}{\pi} \left[I_0 \sin\left(\omega t + \frac{\pi}{4}\right) - I_0 \cos\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$= \frac{\mu_0 \ln 2 I_0}{\pi} \left[\sin\left(\omega t + \frac{\pi}{4}\right) - \cos\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 I_0 \omega \ln 2}{\pi} \left[\cos\left(\omega t + \frac{\pi}{4}\right) + \sin\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$\varepsilon_{\text{max.}} = \frac{\sqrt{2} \mu_0 I_0 \omega \ln 2}{\pi}$$

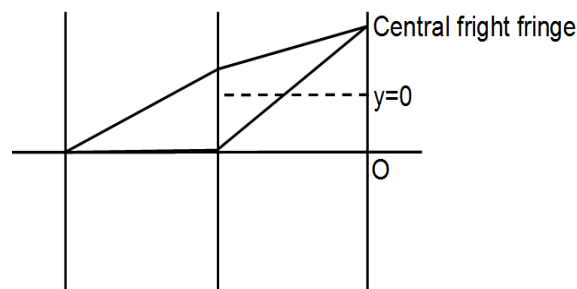
$$Q_{\text{max}} = \frac{\sqrt{2} \mu_0 I_0 \omega \ln 2}{\pi} C$$

$$Q = CE$$

$$\frac{dQ}{dt} = C\varepsilon = C \frac{\mu_0 I_0 \omega^2 \ln 2}{\pi} \left[-\sin\left(\omega t + \frac{\pi}{4}\right) + \cos\left(\omega t + \frac{\pi}{4}\right) \right]$$

Current in frame at $t = \frac{\pi}{\omega}$ is zero.

2. Ans (B,D)



$$\text{Path difference} = SS_1 - SS_2 + S_1O - S_2O$$

$$= \sqrt{D^2 + d^2} - D + \frac{dy}{3D}$$

$$= D \left(1 + \frac{d^2}{2D^2} \right) - D + \frac{dy}{3D}$$

$$\therefore \Delta x = \frac{d^2}{2D} + \frac{dy}{3D}$$

$$\text{At O, } y = \frac{d}{2}$$

$$\therefore \Delta x = \frac{d^2}{2D} + \frac{d^2}{6D} = \lambda$$

$$\therefore d_{\text{min}} = \sqrt{\frac{3\lambda D}{2}}$$

For central maxima,

$$\Delta x = 0$$

$$\therefore y = -\frac{3d}{2}$$

$$\therefore \text{Distance from O is } \frac{d}{2} + \frac{3d}{2}$$

$$= 2d_{\text{min}}$$

$$= \sqrt{6\lambda D}$$

3. Ans (B,C,D)

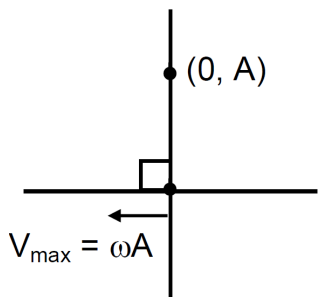
(a) For $(2A, 0)$, $\theta = 0^\circ$ or 180°

$\therefore B = 0$ permanent zero

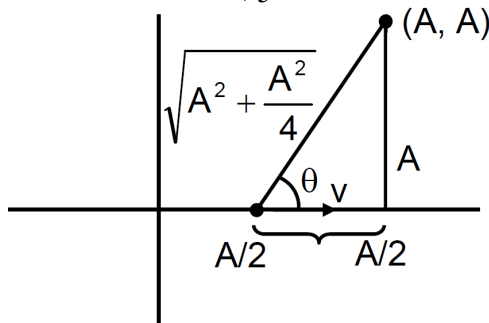
(b) Magnetic field will be max at $(0, A)$ when

the particle passes through $(0, 0)$

$$B_{\max} = \frac{\mu_0}{4\pi} \times \frac{q(\omega A) \sin 90^\circ}{A^2} = \frac{\mu_0}{4\pi} \cdot \frac{q\omega}{A}$$



$$(c) \sin \theta = \frac{A}{A\sqrt{5}/2} = \frac{2}{\sqrt{5}}$$



$$r = \sqrt{A^2 + \frac{A^2}{4}} = \frac{A\sqrt{5}}{2}$$

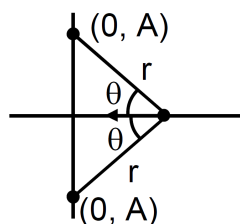
$$V = \omega \sqrt{A^2 - x^2} = \omega \sqrt{A^2 - \left(\frac{A}{2}\right)^2} = \frac{\omega A \sqrt{3}}{2}$$

$$B = \frac{\mu_0}{4\pi} \times \frac{qv \sin \theta}{r^2} = \frac{\mu_0}{4\pi} \times \frac{q \left(\omega A \frac{\sqrt{3}}{2} \right) \cdot \frac{2}{\sqrt{5}}}{\left(\frac{A\sqrt{5}}{2} \right)^2}$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{q\omega}{A} \cdot \frac{\sqrt{3}}{2} \cdot \frac{2}{\sqrt{5}} \times \frac{4}{5} = \frac{\sqrt{3}\mu_0 q \omega}{5\sqrt{5}\pi A}$$

(d) same r , same θ

$\Rightarrow$ same B



4. Ans (B,C,D)

$$h_0 = \frac{h}{2}$$

$$y = \frac{\omega^2 R^2}{2g} \Rightarrow \frac{h}{2} = \frac{\omega^2 R^2}{2g}$$

$$\omega = \sqrt{\frac{gh}{R^2}}$$

$$\frac{dp}{dx} = \rho \omega^2 x$$

$$\frac{dp}{dy} = -\rho g$$

5. Ans (A,C,D)

$$C = \frac{4\pi\epsilon_0 abK}{b-a}$$

$$R = \int_a^b \rho \frac{dr}{4\pi r^2} = \frac{\rho}{4\pi} \frac{(b-a)}{ab}$$

$$\tau = RC = \rho K \epsilon_0$$

$$q = q_0 e^{-t/\tau}$$

$$q = q_0 e^{-\frac{t}{\rho K \epsilon_0}}$$

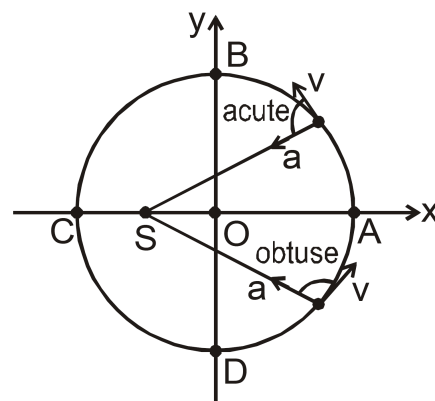
$$t = \infty, q = 0$$

$$\text{Amount of heat generated } H = \left(\frac{1}{a} - \frac{1}{b} \right) \frac{q_0^2}{8\pi\epsilon_0 K}$$

6. Ans (A,C)

As the object moves from A to C via B the angle between acceleration vector and velocity vector decreases from 90° and then increases back to 90° . Since the angle between velocity and acceleration is acute, the object speeds up.

As the object moves from C to A via D the angle between acceleration vector and velocity vector increases from 90° and then decreases back to 90° . Since the angle between velocity and acceleration is obtuse, the object slows down.



PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

$$(A) \vec{k} = 2\hat{i} - 2\hat{j} - \hat{k}$$

$$|\vec{k}| = 3$$

$$\therefore \lambda = \frac{2\pi}{|\vec{k}|} = \frac{2\pi}{3}$$

(D) for angle between wave vector and +ve x-axis

$$\vec{k} \cdot \hat{i} = 2 \Rightarrow (3) (1) \cos\theta = 2$$

$$\Rightarrow \cos\theta = \frac{2}{3}$$

for angle between wave vector and -ve y-axis

$$\vec{k} \cdot (-\hat{j}) = 2 \Rightarrow (3) (1) \cos\theta = 2$$

$$\Rightarrow \cos\theta = \frac{2}{3}$$

8. **Ans (B)**

Component of velocity along the wave vector

$$= \vec{u} \cdot \vec{k} = u\hat{i} \cdot \left(\frac{2\hat{i} - 2\hat{j} - \hat{k}}{3} \right) = \frac{2u}{3}$$

$$\text{Time between time consecutive beat} = \frac{\lambda}{2(\vec{u} \cdot \vec{k})}$$

$$= \frac{\pi}{3 \left(\frac{2u}{3} \right)} = \frac{\pi}{2u} = \frac{1}{2}$$

Frequency = 2 Ans.

9. **Ans (B)**

$$\mu \sin i = 1 \cdot \sin 90^\circ = \text{constant}$$

$$\sin i = \frac{1}{\mu}$$

$$\frac{dy}{dx} = \tan i = \frac{1}{\sqrt{\mu^2 - 1}}$$

$$\frac{dx}{dy} = \cot i = \frac{\sqrt{\mu^2 - 1}}{1} = \sqrt{e^{x/d}} = e^{x/2d}$$

$$\int_0^x e^{-x/2d} dx = \int_0^d dy$$

$$x = 2d \ln 2$$

10. **Ans (A)**

$$\mu = \sqrt{1 + e^{\frac{x}{d}}}$$

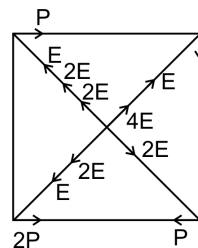
$$\text{Put } x = 2d \ln 2$$

$$\mu = \sqrt{5}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.00)**

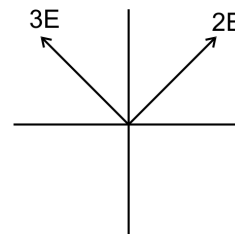
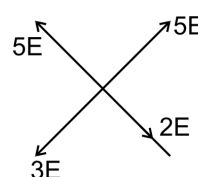


$$E_r = \frac{2KP \cos 45^\circ}{r^3}$$

$$E_r = \frac{2KP}{\sqrt{2}r^3} = 2E$$

$$E_\theta = \frac{KP \sin 45^\circ}{r^3}$$

$$E_\theta = \frac{KP}{\sqrt{2}r^3} = E$$



$$\vec{E}_{\text{net}} = \frac{-3E}{\sqrt{2}} \hat{i} + \frac{3E}{\sqrt{2}} \hat{j} + \frac{2E}{\sqrt{2}} \hat{i} + \frac{2E}{\sqrt{2}} \hat{j}$$

$$\vec{E} = -\frac{E}{\sqrt{2}} \hat{i} + \frac{5E}{\sqrt{2}} \hat{j}$$

$$= -\frac{1}{\sqrt{2}} \frac{KP}{\sqrt{2}r^3} \hat{i} + \frac{5}{\sqrt{2}} \frac{KP}{\sqrt{2}r^3} \hat{j}$$

$$= \frac{KP}{r^3} \left(-\frac{1}{2} \hat{i} + \frac{5}{2} \hat{j} \right)$$

$$= \frac{KP}{r^3} (x\hat{i} + y\hat{j})$$

$$\therefore x + y = -\frac{1}{2} + \frac{5}{2} = \frac{4}{2} = 2$$

2. **Ans (8.00)**

$$U_1 = \frac{1}{2} Li^2 = \frac{1}{2} \cdot L \cdot \left[\frac{\epsilon}{R} (1 - e^{-1}) \right]^2$$

$$U_1 = \frac{1}{2} L \cdot \frac{\epsilon^2}{R^2} \left(1 - \frac{1}{e} \right)^2 \quad \dots (i)$$

$$U_2 = \int i^2 R dt = R \cdot \frac{\epsilon^2}{R^2} \int \left(1 - e^{-\frac{t}{\tau}} \right)^2 dt$$

$$U_2 = \frac{\epsilon^2 L}{R^2} \left[\frac{2}{e} - \frac{1}{2e^2} - \frac{1}{2} \right]$$

$$\frac{U_1}{U_2} = \frac{1}{2} \frac{\left(1 - \frac{1}{e} \right)^2}{\left(\frac{2}{e} - \frac{1}{2e^2} - \frac{1}{2} \right)}$$

3. Ans (9.00)

Let ℓ_1 and ℓ_2 be the final length of the two parts, then from gas equation

$$\frac{P_0 A \ell_0}{T_0} = \frac{P A \ell_1}{T_1} = \frac{P A \ell_2}{T_2} \quad \dots\dots (i)$$

Considering the equilibrium of the piston in initial and final states, we get

$$P_0 A = k x_0, PA = kx.$$

$$\Rightarrow \frac{P}{P_0} = \frac{x}{x_0} \quad \dots\dots(ii)$$

decrease in the length of spring = total increase in the length of the two chambers

$$x - x_0 = \ell_1 + \ell_2 - 2\ell_0 \quad \dots\dots (iii)$$

from relation (i)

$$\ell_1 = \frac{P_0 T_1 \ell_0}{P T_0}, \ell_2 = \frac{P_0 T_2 \ell_0}{P T_0}$$

using (ii)

$$\ell_1 = \frac{x_0 T_1 \ell_0}{x T_0}, \ell_2 = \frac{x_0 T_2 \ell_0}{x T_0}$$

putting these in (iii)

$$x - x_0 = \frac{x_0 \ell_0}{x T_0} (T_1 + T_2) - 2\ell_0$$

$$\text{putting values and solving, we get } x = \frac{3}{20} m$$

4. Ans (6.00)

$$\text{compression in rod A} = \frac{mg}{3}; \text{ young modulus} = Y_A$$

$$\text{compression in another rod} = \frac{2mg}{3}; \text{ young modulus} = Y_B$$

Since compression in both rod is same

$$\frac{\frac{mg}{3}}{Y_A} = \frac{\frac{2mg}{3}}{Y_B} \Rightarrow \frac{Y_B}{Y_A} \Rightarrow 2$$

Given

$$Y_A \alpha_A \Delta T = 3 Y_B \alpha_B \Delta T$$

$$\frac{\alpha_A}{\alpha_B} = \frac{3 Y_B}{Y_A} = 6$$

5. Ans (2.00)

$$\epsilon_i = \epsilon_f$$

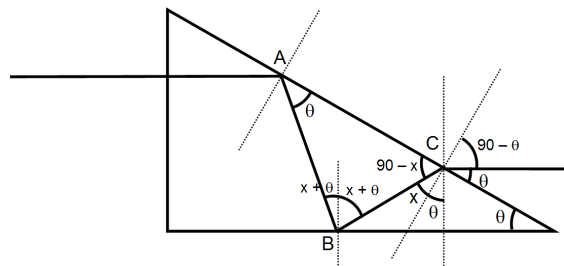
$$mg \frac{L}{2} \sin \theta + 0 = m_1 g \frac{(L-x)}{2} \sin \theta - mg \frac{x}{2} + \frac{1}{2} m v^2$$

$$mg \frac{L}{2} \sin \theta = \frac{g}{L} (L^2 + x^2 - 2Lx) \sin \theta - \frac{x^2}{L} + v^2$$

$$gL \sin \theta = \frac{g}{L} (L^2 + x^2 - 2Lx) \sin \theta - \frac{x^2}{L} + v^2$$

$$v = \sqrt{\frac{5}{8} g \ell}$$

6. Ans (0.75)



In $\triangle ABC$,

$$\theta + 90 - x + 2(x + \theta) = 180$$

$$\therefore x = 90 - 3\theta$$

$$\Rightarrow \mu \sin x = \sin(90 - \theta)$$

$$\therefore \mu = \frac{\cos \theta}{\cos 3\theta}$$

$$\mu = \frac{\cos 15^\circ}{\cos 45^\circ} = \frac{\frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{1}{\sqrt{2}}}$$

$$= \sqrt{\frac{3}{4}} + \frac{1}{2}$$

$$\therefore n = 0.75$$

7. Ans (7.00)

Let N_0 be the initial number of nuclei, then

$$N_1 = N_0 e^{-\lambda t_1}$$

$$\text{and } N_2 = N_0 e^{-\lambda t_2}$$

$$\therefore \text{number of nuclei decayed} = N_1 - N_2$$

$$= N_0 (e^{-\lambda t_1} - e^{-\lambda t_2})$$

$$= \frac{A_0}{\lambda} (e^{-\lambda t_1} - e^{-\lambda t_2})$$

$$= \frac{A_1 - A_2}{\lambda} = (A_1 - A_2) \tau.$$

8. Ans (6.04)

$$\text{Zero error} = 4 \times 0.005 = 0.02 \text{ mm}$$

$$LC = \frac{\text{Pitch}}{\text{Total CSD}}$$

$$\text{Pitch} = 0.005 \times 200 = 1 \text{ mm}$$

$$\text{Measured diameter} = (6 \times \text{MSD}) + (12 \times 0.005)$$

$$= (6 \times 1 \text{ mm}) + 0.06$$

$$\text{Measured value} = 6.06 \text{ mm}$$

$$\text{Actual diameter} = 6.06 - e$$

$$= 6.06 - 0.02$$

$$= 6.04 \text{ mm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)

In complexes A and B, one halide (Cl^- or Br^-) is outside the coordination sphere and complexes are :

$[\text{Cr}(\text{NH}_3)_4\text{Br}_2]\text{Cl}$ and $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Br}$ 'A' gives white precipitate of AgCl with excess of AgNO_3 which dissolve in excess ammonia. Therefore, A must be $[\text{Cr}(\text{NH}_3)_4\text{Br}_2]\text{Cl}$.

B gives a pale yellow precipitate with excess of AgNO_3 , which dissolve in concentrated ammonia solution. Therefore, precipitate is AgBr and complex B is $[\text{Cr}(\text{NH}_3)_4\text{ClBr}]\text{Br}$. In both A and B, hybridisation of chromium is d^2sp^3 and magnetic moment ($\mu = \sqrt{n(n+2)} \text{ BM}$) is 0.

2. Ans (B)

Degree of dissociation $\% \alpha = \frac{x}{a} \times 100$

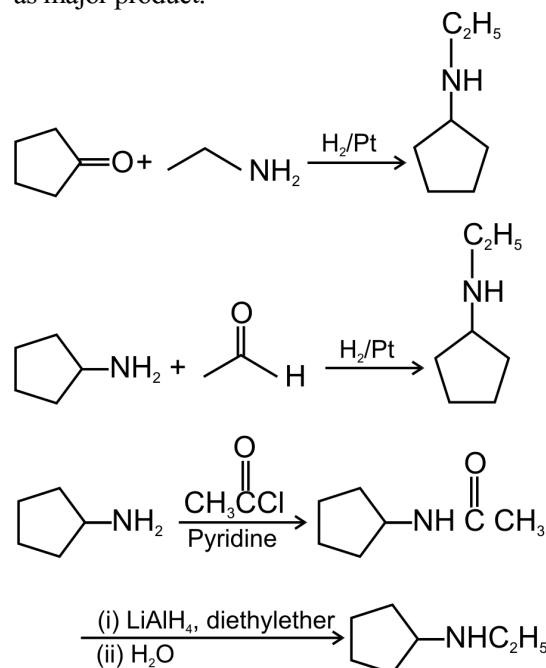
$$\% \alpha = \frac{4}{10} \times 100 = 40\%$$

3. Ans (A,B,C)

Li^+ ion being the smallest in size and has the highest charge/size ratio amongst the alkali metal ions, get much more hydrated (i.e. holds more water molecules in its hydration sphere) than Na^+ ion and the latter gets more hydrated than K^+ ion and so on.

5. Ans (A,B,C)

Reaction (a, b, c) all give N-ethylcyclopentylamine as major product.



6. Ans (A,B,C)

$\Delta S (x \rightarrow y)$ taking place at constant temperature.

$$\begin{aligned} \therefore \Delta S &= 2.303nR \log \frac{V_2}{V_1} \\ &= 2.303nR \log \frac{p_1}{p_2} \\ &= 2.303 \times 5 \times 8.314 \log \left(\frac{10}{1} \right) \\ &= 95.74 \text{ JK}^{-1} \end{aligned}$$

Thus, (a) is the correct option.

(b) $\because q = 0$,

$$\therefore \Delta S (x \rightarrow z) = \frac{q}{T} = 0$$

Thus, (b) is the correct option.

(c) $x \rightarrow z$ takes place adiabatically, hence temperature falls to 118.6 K.

$$\Delta S_{x \rightarrow z} = 0, C_p = \frac{5}{2} R;$$

For monoatomic gas

$$\begin{aligned} \Delta S_{z \rightarrow y} &= 2.303n C_p \log \frac{T_2}{T_1} \\ &= 2.303 \times 5 \times 2.5 \times 8.314 \log \frac{298}{118.6} \\ &= 95.76 \text{ J K}^{-1} \end{aligned}$$

$$\begin{aligned} \text{Thus, } \Delta S_{x \rightarrow y} &= \Delta S_{x \rightarrow z} + \Delta S_{z \rightarrow y} \\ &= 95.76 \text{ J K}^{-1} \end{aligned}$$

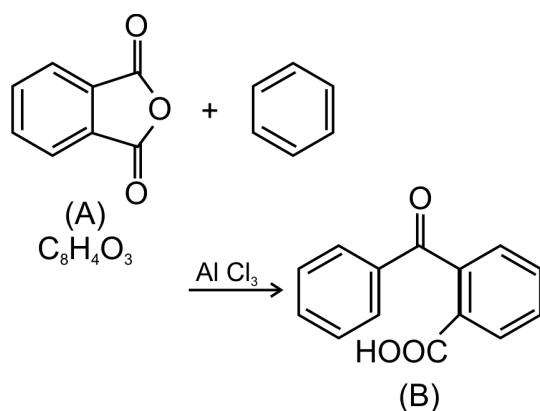
$\therefore$ (c) is the correct option

(d) $\Delta S_{x \rightarrow z} \rightarrow y \neq 0$, thus (d) is the incorrect option.

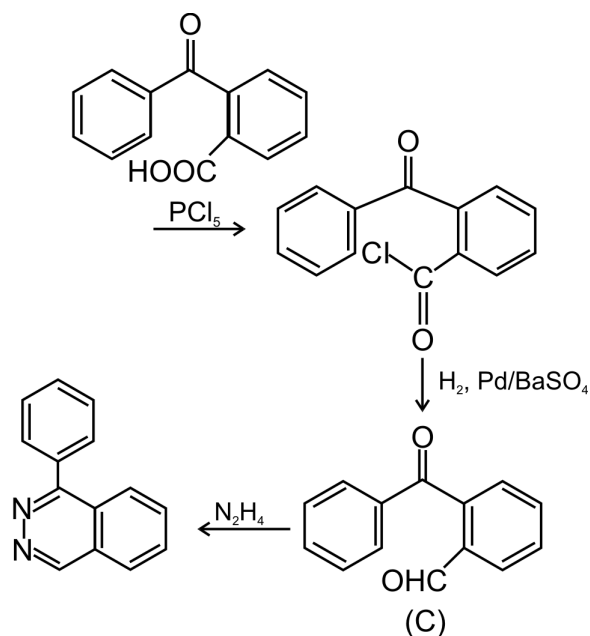
PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (B)**



8. **Ans (C)**



9. **Ans (A)**

$$\Delta U = nC_{v,m}\Delta T$$

$$\Delta U = 2 \times \frac{3}{2} \times R \times T_A$$

10. **Ans (B)**

$$W_{AB} = -nR\Delta T$$

$$= -2 \times R \times T_A$$

$$= -2RT_A$$

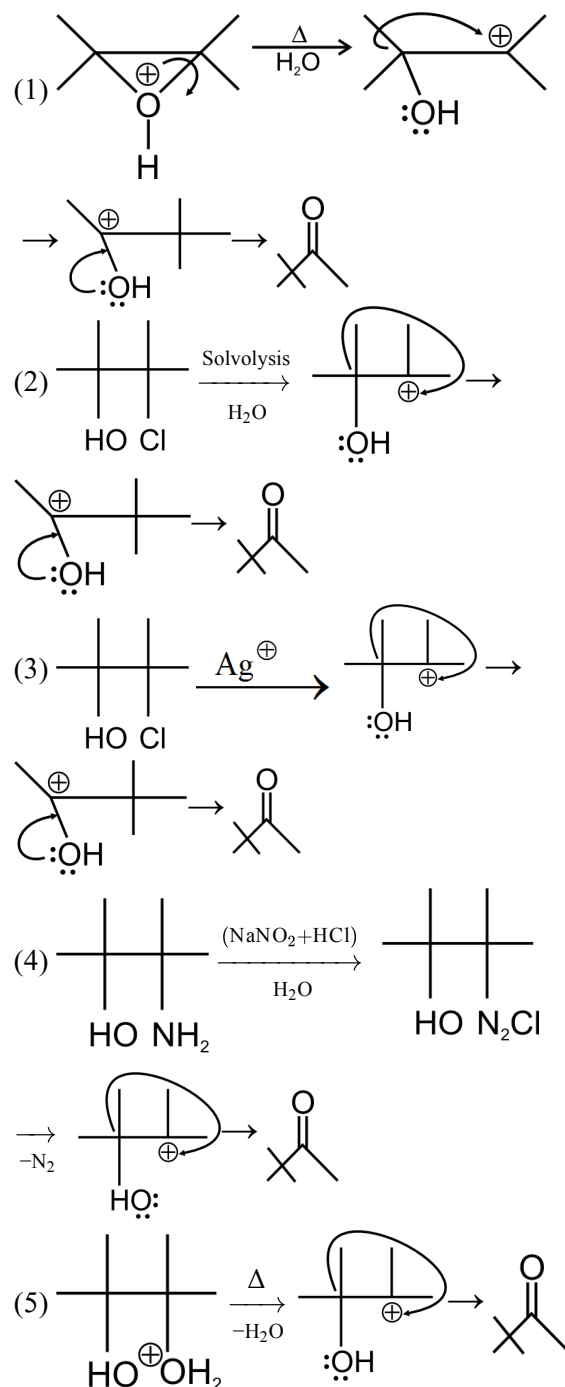
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5.00)**

II group radicals Pb^{2+} , Cu^{2+} , Cd^{2+} , Hg^{2+} , Sn^{2+} are precipitated with H_2S in acidic medium. IV group cations Zn^{2+} , Mn^{2+} , Co^{2+} , Ni^{2+} are precipitated with H_2S in alkaline medium.

2. **Ans (5.00)**



3. Ans (5.33)

$$\text{Moles of metal} = \frac{pV}{RT} = \frac{6.08}{760} \times \frac{0.5}{0.08 \times 1000}$$

$$= 9.33 \times 10^{-5}$$

$$\Rightarrow \text{Total number of atoms} = 9.33 \times 10^{-5} \times 6.023 \times 10^{23}$$

$$= 5.62 \times 10^{19} \text{ atoms}$$

$\therefore$ In bcc, there are 2 atoms per unit cell.

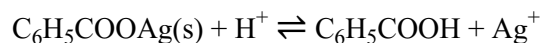
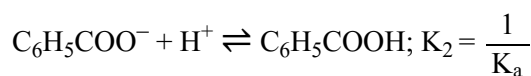
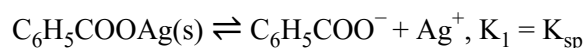
$\Rightarrow$ Number of unit cells in given sample

$$= \frac{1}{2} \times 5.62 \times 10^{19} = 2.81 \times 10^{19}$$

$$\Rightarrow \text{Volume of one unit cell} = \frac{(1.62 \times 0.1)^3}{2.81 \times 10^{19}}$$

$$= 5.33 \times 10^{-22} \text{ cm}^3$$

4. Ans (10.00)



$$K_3 = \frac{K_{sp}}{K_a}$$

Let x is the solubility of $\text{C}_6\text{H}_5\text{COOAg}$ in a buffer of pH = 3.19

$$\therefore [\text{H}^+] = 10^{-3.19} = 6.46 \times 10^{-4} \text{ M}$$

$$\therefore K_3 = \frac{[\text{C}_6\text{H}_5\text{COOH}][\text{Ag}^+]}{[\text{H}^+]} = \frac{x \cdot x}{[\text{H}^+]} = \frac{x^2}{[\text{H}^+]} = \frac{K_{sp}}{K_a}$$

$$\therefore x = \sqrt{\frac{2.5 \times 10^{-13}}{6.46 \times 10^{-5}} \times (6.46 \times 10^{-4})}$$

$$x = 1.6 \times 10^{-6} \text{ M}$$

In aqueous solution, solubility

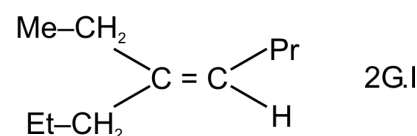
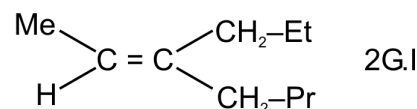
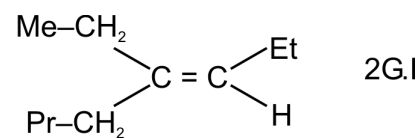
$$S = \sqrt{2.5 \times 10^{-13}} = 5 \times 10^{-7} \text{ M}$$

$$\therefore \frac{x}{5} = \frac{1.6 \times 10^{-6}}{5 \times 10^{-7}} = 3.20$$

5. Ans (6.00)

Three different β -hydrogens are present in the reactant

$\therefore$ 5 different products will be form



6. Ans (9.00)

$$P_T = X_A P_A^\circ + X_B P_B^\circ$$

$$320 = X_A \times 350 + (1 - X_A) \times 200$$

$$X_A = 0.8$$

$$X_A P_A^\circ = Y_A P_T$$

$$Y_A = \frac{0.8 \times 350}{320}$$

$$Y_A = \frac{7}{8}$$

Number of moles of ethylalcohol in vapour

$$\text{phase} = \frac{7}{8} \times 10.29 = 9$$

7. Ans (5.00)

Millimoles of oxalic acid absorbed

$$= 40 \times 0.5 - 0.25 \times 40 = 10$$

$$\therefore \text{Number of moles of oxalic acid} = 10 \times 10^{-3}$$

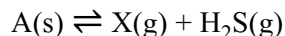
$$\text{amount of oxalic acid} = 10 \times 10^{-3} \times 126 \text{ g}$$

$$= 1.26 \text{ g}$$

$\therefore$ Amount of oxalic acid adsorbed by per gram of charcoal

$$= \frac{1.26}{25.2} = 5 \times 10^{-2} = 0.05$$

8. Ans (84.40)

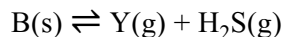


$$K_A = p \times p_{H_2S}$$

$$\therefore px = p_{H_2S} = 34 \text{ mm } [\because px = p_{H_2S}]$$

$$\therefore K_A = (34 \times 34) = 1156 \text{ mm}^2$$

Similarly,



$$K_B = py \times p_{H_2S}$$

$$\therefore py + p_{H_2S} = 50 \text{ mm}$$

$$\therefore py = p_{H_2S} = 25 \text{ mm } [\because py = p_{H_2S}]$$

$$\Rightarrow K_B = 625.2 \text{ mm}$$

In the mixture let, $px = x$, $py = y$, then total

$$p_{H_2S} = (x + y)$$

$$\therefore K_A = px \cdot p_{H_2S} = x(x + y) = 1156 \quad \dots(i)$$

$$K_B = py \cdot p_{H_2S} = (x + y) = 625 \quad \dots(ii)$$

$$\therefore \frac{x}{y} = \frac{K_A}{K_B} = 1.85 \Rightarrow x = 1.85 y$$

On putting value of x in Eq. (i),

$$\text{we have } 1.85y(1.85y + y) = 1156$$

$$\Rightarrow y = 14.81 \text{ mm and } x = 27.40 \text{ mm}$$

$$\therefore \text{Total pressure} = px + py + p_{H_2S}$$

$$= 2(x + y)$$

$$= 2(14.81 + 27.40)$$

$$= 84.42 \text{ mm}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C,D)

We

have,

$$\cot^{-1} \frac{1}{x} = \begin{cases} \pi + \tan^{-1} x, & x < 0 \\ \tan^{-1} x, & x > 0 \end{cases} \text{ and } \tan^{-1} x +$$

$$\cot^{-1} x = \frac{\pi}{2} \quad \forall x \in \mathbb{R}$$

$$\text{Also } \frac{|\sin x|}{\sin x} \text{ is } \begin{cases} 1 & \sin x > 0 \\ -1 & \sin x < 0 \end{cases}$$

$$\text{also period of } \tan^4(ax) + \cot^4(ax) \text{ is } \pi/2a$$

2. Ans (A,B)

Take any four number

$$\text{Ex : } 5, 1, 3, 3$$

They can be arranged only in one way such that each number is not greater than the preceding

$$\text{i.e. } 5, 3, 3, 1$$

$$\text{Probability} = \frac{{}^5C_1 \cdot 1 + {}^5C_2 \cdot 2 + {}^5C_3 \cdot 1}{5^3} = \frac{35}{125}$$

3. Ans (C,D)

$$A(2t_1, t_1^2) B(2t_2, t_2^2)$$

$$t_1 t_2 = -4$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 2t_1 & t_1^2 & 1 \\ 2t_2 & t_2^2 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 20$$

$$\text{solving } t_1 = \pm 4, \pm 1$$

4. Ans (A,B,C)

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} a-\lambda & b & c \\ d & e-\lambda & f \\ g & h & i-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda^3 + \lambda^2(a + e + i) - \lambda(ae + ei + ia - gc - hf - bd) +$$

$$|A| = 0$$

Using Cayley Hamilton,

$$A^3 - A^2 \text{tr}(A) + \frac{(\text{tr } A)^2 - \text{tr } A^2}{2} A - |A| \cdot I_3 = 0$$

5. Ans (A,B,C)

Let $a_1 = 5 - d$, $a_2 = 5$, $a_3 = 5 + d$, $b_1 = \frac{3}{q}$,
 $b_2 = 3$, $b_3 = 3q$, then the given conditions
indicate that $5 - d + \frac{3}{q}$ and $5 + d + 3q$ are all
positive integers and

$$\left(5 - d + \frac{3}{q}\right)(5 + d + 3q) = 64.$$

It is easy to check that for getting maximum
positive d , there are only four possibilities for
 $\left(5 - d + \frac{3}{q}, 5 + d + 3q\right) : (1, 64), (2, 32), (4, 16)$
and $(8, 8)$.

(i) By solving the system of equations
corresponding to $(1, 64)$, it follows that

$$3q + \frac{3}{q} = 55 \Rightarrow q_{1,2} = \frac{55 \pm 7\sqrt{61}}{6}, d_{1,2} = 4 + \frac{3}{q_{1,2}}$$

$$\Rightarrow d_{\max} = 4 + \frac{3.6}{55 - 7\sqrt{61}} = 4 + \frac{55 + 7\sqrt{61}}{2}$$

(ii) By solving the system of equations
corresponding to $(2, 32)$, it follows that

$$q + \frac{1}{q} = 8 \Rightarrow q_{1,2} = \frac{8 \pm \sqrt{60}}{2} = 4 \pm \sqrt{15}, d_{1,2} = 3 + \frac{3}{q_{1,2}}$$

$$\Rightarrow d_{\max} = 3 + \frac{3}{4 - \sqrt{15}} = 15 + 3\sqrt{15}.$$

(iii) By solving the system of equation
corresponding to $(4, 16)$, it follows that

$$3q + \frac{3}{q} = 10 \Rightarrow q = 3 \text{ or } \frac{1}{3}, d_{1,2} = 1 + \frac{3}{q_{1,2}} = 2$$

$$\text{or } 10 \Rightarrow d_{\max} = 10$$

(iv) By solving the system of equations
corresponding to $(8, 8)$, it follows that

$$q + \frac{1}{q} = 2 \Rightarrow q = 1, d = 0.$$

In summary, $d_{\max} = 4 + \frac{55 + 7\sqrt{61}}{2}$, hence the
maximum possible value of a_3 is $9 + \frac{55 + 7\sqrt{61}}{2}$

6. Ans (B,C,D)

$$g(x) = \begin{cases} f(x) & x > 0 \\ -f(x) & x < 0 \end{cases}$$

$$h(x) = \tan(\cos^{-1}(\sin x))$$

$$h(x) = \tan\left(\frac{\pi}{2} - \sin^{-1}(\sin x)\right)$$

$$= \tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$h(x) = \cot x$$

$$h(f(x)) = \cot(f(x))$$

$$\therefore f(x) = \tan^{-1}\left(\frac{2}{x}\right)$$

$$\tan(f(x)) = \left(\frac{2}{x}\right)$$

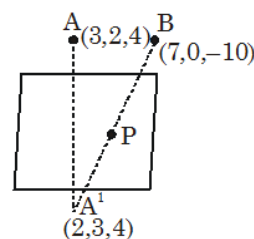
$$\cot(f(x)) = \left(\frac{x}{2}\right)$$

Now check options

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)

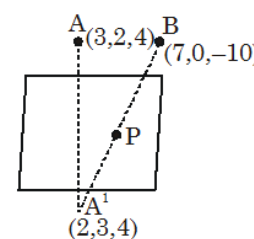


A + B lie on same side of plane, for
 $|PA + PB|_{\min}$ P must be point of intersects
of line A'B with plane where A' is mirror image
of A

$$\text{Line } A'B \Rightarrow \frac{x-2}{5} = \frac{y-3}{-3} = \frac{z-4}{-14} = \lambda$$

$$\Rightarrow P\left(\frac{21}{8}, \frac{21}{8}, \frac{9}{4}\right)$$

8. Ans (A)



$$\vec{AB} = 2(2\hat{i} - \hat{j} - 7\hat{k}); \vec{n} = \hat{i} - \hat{j}$$

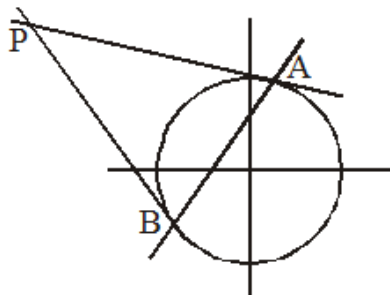
$$\cos \theta = \frac{\vec{AB} \cdot \vec{n}}{|\vec{AB}| |\vec{n}|} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{2\sqrt{3}}\right)$$

9. **Ans (B)**

$$C_1 : x^2 + y^2 = 4$$

$$C_2 : y = x + 1$$

while region R is the disc centered at $(\sqrt{3}, 1)$ and radius = a



Let $P(x_1, y_1)$

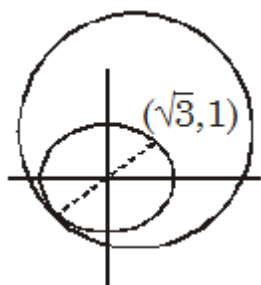
$$\text{Clearly } AB : y - x = 1 \quad \dots\dots(1)$$

is chord of contact

$$\therefore AB : xx_1 + yy_1 = 4 \quad \dots\dots(2)$$

(1) & (2) are identical

$$\Rightarrow (x_1 y_1) = (-4, 4)$$

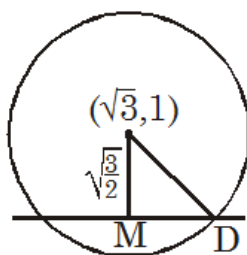


10. **Ans (C)**

Smallest possible value of 'a' for which

$$A \cap C = A \text{ is } a = 4.$$

$\therefore$



$$R : (x - \sqrt{3})^2 + (y - 1)^2 \leq 4^2$$

$$CD : y = x + 1$$

$$P = \left| \frac{\sqrt{3} + 1 - 1}{\sqrt{2}} \right| = \sqrt{\frac{3}{2}}$$

$$MD = \sqrt{16 - \frac{3}{2}} = \sqrt{\frac{29}{2}}$$

$$\therefore CD = \sqrt{58}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (4.00)**

$$1 = \lim_{x \rightarrow \infty} x \ln \left(\frac{e(1 + (1/x))}{(1 + (1/x))^x} \right) \quad (\infty \times 0 \text{ form})$$

$$= \lim_{x \rightarrow \infty} x \left(1 + \ln \left(1 + \frac{1}{x} \right) - x \ln \left(1 + \frac{1}{x} \right) \right)$$

$$\text{put } x = \frac{1}{t}; \text{ as } x \rightarrow \infty, t \rightarrow 0$$

$$\text{Hence } 1 = \lim_{t \rightarrow 0} \frac{1}{t} \left(1 + \ln(1+t) - \frac{\ln(1+t)}{t} \right)$$

$$= \lim_{t \rightarrow 0} \left[\ln(1+t)^{1/t} + \frac{t - \ln(1+t)}{t^2} \right]$$

$$= 1 + \lim_{y \rightarrow 0} \left(\frac{e^y - 1 - y}{y^2} \right) \text{ where } \ln(1+t) = y;$$

$$1+t = e^y, \text{ hence } t = e^y - 1$$

$$= 1 + \frac{1}{2} = \frac{3}{2} = \frac{m}{n}$$

2. **Ans (3.00)**

We have,

$$S_{100} = \frac{0}{({}^{100}C_0)^5} + \frac{1}{({}^{100}C_1)^5} + \frac{2}{({}^{100}C_2)^5} + \dots + \frac{100}{({}^{100}C_{100})^5} \quad \dots\dots(1)$$

Also,

$$S_{100} = \frac{100}{({}^{100}C_0)^5} + \frac{(100-1)}{({}^{100}C_1)^5} + \frac{(100-2)}{({}^{100}C_2)^5} + \dots + \frac{0}{({}^{100}C_{100})^5} \quad \dots\dots(2) \left(\sum_{r=a}^b f(r) = \sum_{r=a}^b f(a+b-r) \right)$$

$\therefore$ On adding (1) and (2), we get

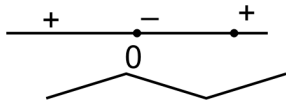
$$2S_{100} = 100 t_{100} + 100 \Rightarrow S_{100} - 50 = 50 t_{100}$$

$$\frac{S_{100} - 50}{150 t_{100}} = \frac{1}{3}$$

$$\text{Hence, } \sec \left(\cos^{-1} \left(\frac{S_{100} - 50}{150 t_{100}} \right) \right) = 3$$

3. Ans (1.00)

$$f(x) = x^2 - 3x$$



$$f(x) = \frac{x^3}{3} - \frac{3}{2}x^2 + c$$

$$f(1) = \frac{5}{6}$$

$$\therefore \frac{1}{3} - \frac{3}{2} + c = \frac{5}{6}$$

$$\Rightarrow c = 2$$

$$f(x) = \frac{x^3}{3} - \frac{3}{2}x^2 + 2$$

Minimum ordinate will be at $x = 3$

$$\therefore f(3) = 9 - \frac{27}{2} + 2 = -\frac{5}{2}$$

$$\therefore \text{point } (a, b) = \left(3, -\frac{5}{2}\right)$$

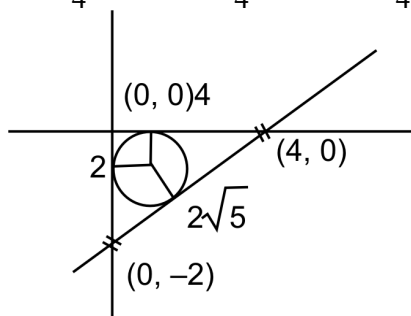
$$2(a + b) = 1$$

4. Ans (3.00)

$$x^2y - 2xy^2 - 4xy = 0 \Rightarrow xy(x - 2y - 4) = 0$$

$$r = \frac{\Delta}{s} = \frac{\frac{1}{2} \times 2 \times 4}{3 + \sqrt{5}} = \frac{4}{3 + \sqrt{5}}$$

$$= \frac{2}{r} = \frac{2(3 + \sqrt{5})}{4} = \frac{6 + 2\sqrt{5}}{4} = \frac{(\sqrt{5} + 1)^2}{4}$$



$$\frac{2}{r} = 4\cos^2 \frac{\pi}{5}$$

$$= \frac{1}{r} = 2\cos^2 \frac{\pi}{5}$$

5. Ans (3.00)

$$f(x) = (x - 1)(x - 2)(x - 3)(c - x) + x^4 - x$$

coefficient of $x^3 = 1$

$$\therefore c + 1 + 2 + 3 = 1 \Rightarrow c = -5$$

$$f(x) = (x - 1)(x - 2)(x - 3)(-5 - x) + x^4 - x$$

$$f(4) = 3 \cdot 2 \cdot 1 \cdot (-9) + 4^4 - 4$$

$$= 198 = 2 \times 3^2 \times 11$$

6. Ans (2.00)

$$P(x) - P(x + 1) = 2x - 1 \quad \forall x \in \mathbb{R}$$

$$P(x) = ax^2 + bx + c$$

$$ax^2 + bx + c - [a(x + 1)^2 + b(x + 1) + c] = 2x - 1$$

$$ax^2 + bx - [ax^2 + 2ax + a + bx + b] = 2x - 1$$

$$-2ax - (a + b) = 2x - 1$$

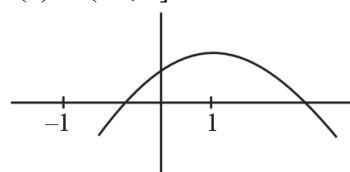
$$-2a = 2 \quad a + b = 1$$

$$a = -1 \quad b = 2$$

$$P(x) = -x^2 + 2x + c$$

$\therefore$ Greatest value of

$P(x)$ in $(-\infty, 0]$ is 2



$$\therefore P(0) = c = 2$$

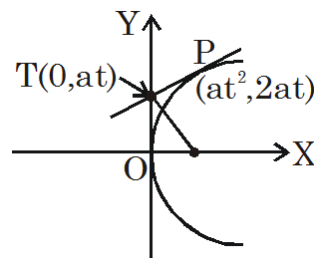
$$\therefore P(x) = -x^2 + 2x + 2$$

$$P(1) = -1 + 2 + 2 = 3$$

$$P(-1) = -1 - 2 + 2 = -1$$

$$P(1) + P(-1) = 2$$

7. Ans (9.00)



(h, k) is mid point of PT then

$$h = \frac{at^2}{2}; \quad k = \frac{2at + at}{2}$$

$$\Rightarrow 2h = a \cdot \frac{4k^2}{9a^2}$$

$$\Rightarrow 2y^2 = 9ax$$

$$a = 2$$

$$y^2 = 9x$$

$$L.R = 9$$

8. Ans (1.20)

$$\frac{f(5) - f(3)}{5 - 3} = f'(c) \quad (\text{By LMVT})$$

$$\therefore 1 < \frac{f(5) - 1}{2} < 2$$

$$\Rightarrow 3 < f(5) < 5 \therefore \lambda = 3, \mu = 5$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

16-04-2023

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,C,D	A,C	B,C,D	A,C,D	A		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	2.00	0.31	0.25	7.00	3.00	1.60	5.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	D	A,C	B,D	A,B,C,D	A,B,D	B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	6.00	250.00	0.00	5.00	5.00	5.00	5.00	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	B,C,D	A,C	B,C,D	A,B,C	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	B	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	5.00	1356.00	4.00	2.00	5.00	5.00	28.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

Process AB is isothermal so $T_A = T_B = T_0$

$$\text{Now } \Delta U_{BC} = \frac{nfR(T_C - T_B)}{2}$$

{we need to calculate T_C }

Process CA is adiabatic

$$\Rightarrow T_C \cdot V_C^{\gamma-1} = T_A \cdot V_A^{\gamma-1}$$

$$\Rightarrow T_C \cdot (2V_0)^{5/3-1} = T_0(V_0)^{5/3-1}$$

$$\Rightarrow T_C = \frac{T_0}{2^{2/3}} = \frac{2T_0}{3}$$

$$\text{Now } \Delta U_{BC} = \frac{1 \times 3 \times R \left(\frac{2T_0}{3} - T_0 \right)}{2} = \frac{-RT_0}{2}$$

Option A is correct

→ Work done for all processes

$$AB \rightarrow W_{AB} = nRT_0 \ln \left(\frac{2V_0}{V_0} \right)$$

$$= 1 \times RT_0 \ln 2 = 0.7RT_0$$

$$\Delta U_{AB} = 0 \text{ \& } Q_{AB} = \Delta U_{AB} + W_{AB} = 0.7RT_0$$

$$BC \rightarrow W_{BC} = 0, \Delta U_{BC} = -RT_0/2$$

$$Q_{BC} = W_{BC} + \Delta U_{BC} = -RT_0/2$$

$$CA \rightarrow W_{CA} = -\Delta U_{CA}$$

$$= -\frac{1 \times 3 \times R \left(T_0 - \frac{2T_0}{3} \right)}{2} = -\frac{RT_0}{2}$$

No heat is supplied in CA

Now net work done

$$\Rightarrow W = W_{AB} + W_{BC} + W_{CA}$$

$$= 0.7RT_0 + 0 - 0.5RT_0$$

$$\Rightarrow W = 0.2RT_0 = RT_0/5$$

Heat is supplied to gas only in process AB that is $0.7RT_0$

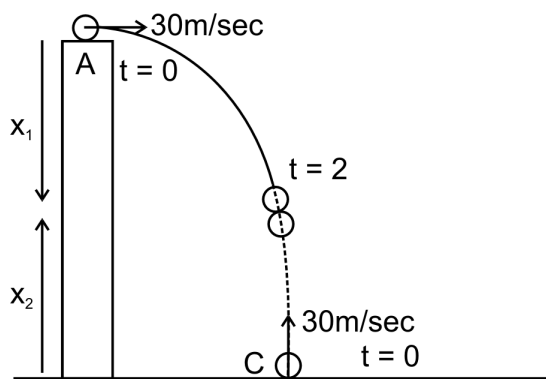
$$\text{Now efficiency} = \frac{\text{Work Done}}{\text{Heat Supplied}}$$

$$= \frac{2RT_0}{7RT_0} = \frac{2}{7}$$

2. Ans (A, C, D)

As both particle strike after 2sec.

Particle-1



Particle-2

$$\text{Distance } x_1 = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 2^2 = 20$$

$$\text{Distance } x_2 = 30t - \frac{1}{2}gt^2$$

$$x_2 = 60 - \frac{1}{2} \times 10 \times 4 = 40$$

$$\Rightarrow \text{Height AB} = x_1 + x_2 = 60 \text{ meter}$$

Distance BC = horizontal distance travelled by particle 1

$$BC = 30 \times 2 = 60 \text{ meter}$$

Option A is correct.

Now after collision particle are combined so collision is inelastic. Applying momentum conservation

$$\vec{p}_i = \vec{p}_f$$

$$\Rightarrow m\vec{v}_1 + 2m\vec{v}_2 = (m + 2m)\vec{v}$$

$$m(30\hat{i} - 20\hat{j}) + 2m(30 - 10 \times 2)\hat{j} = 3m\vec{v}$$

$$m(30\hat{i} - 20\hat{j}) + 20m\hat{j} = 3m\vec{v}$$

$$\Rightarrow 30m\hat{i} = 3m\vec{v} \Rightarrow \vec{v} = 10\hat{i}$$

Therefore after collision combined mass move with speed 10m/sec in horizontal direction.

Option B is wrong

After this, combined mass travel horizontally & reaches point D

Now, CD = range

$$= 10\sqrt{\frac{2 \times H}{g}} = 10\sqrt{\frac{2 \times 40}{10}}$$

$$= 10\sqrt{8} = 20\sqrt{2} = 20 \times 1.4 = 28$$

$$BD = BC + CD = 60 + 28 = 88 \text{ meter}$$

Option C is correct

Time of flight of a particle is the time for which it remains in air

$\Rightarrow$ Time of flight of

$$A = 2 + \sqrt{\frac{2 \times H}{g}} = 2 + \sqrt{\frac{2 \times 40}{10}}$$

$$A = 2 + \sqrt{8} = 2 + 2\sqrt{2}$$

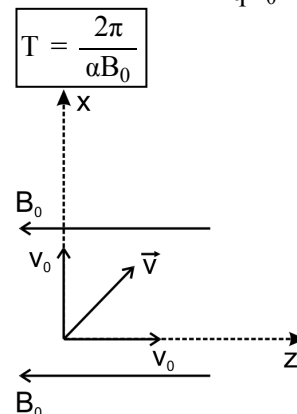
$$A = 2 + 2(1.4) = 4.8 \text{ sec.}$$

3. Ans (A, C)

$$\vec{v} = v_0\hat{i} + v_0\hat{k}$$

Particle will move on helical path.

$$\text{Time period } T = \frac{2\pi m}{qB_0}$$



$$(A) t = \frac{\pi}{\alpha B_0} = \frac{T}{2}$$

After this time component $\perp$ to $\vec{B}$ will change by 180°

$$\text{So } \vec{v}_{t=T/2} = -v_0\hat{i} + v_0\hat{k} = -v_0(\hat{i} - \hat{k})$$

(A) is correct.

(B) In magnetic field, speed of the particle does not change initial speed $= \sqrt{v_0^2 + v_0^2} = \sqrt{2}v_0$

(B) is wrong

(C) Distance = speed $\times$ time

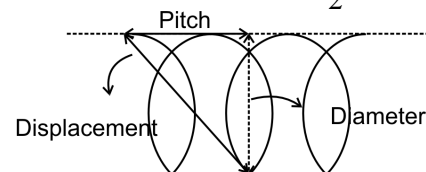
$$= \sqrt{2}v_0 \left(\frac{2\pi}{\alpha B_0} \right) = \frac{2\sqrt{2}v_0\pi}{\alpha B_0}$$

(C) is correct.

$$(D) t = \frac{3\pi}{\alpha B_0} = nT$$

$$\Rightarrow \frac{3\pi}{\alpha B_0} = n \cdot \frac{2\pi}{\alpha B_0} \Rightarrow n = \frac{3}{2}$$

So particle will complete $\frac{3}{2}$ rotation



$$\text{Pitch} = v_0 \cdot t = v_0 \cdot \frac{3\pi}{\alpha B_0} = \frac{3\pi v_0}{\alpha B_0}$$

$$\text{Diameter} = 2(\text{radius}) = 2 \left(\frac{mv_0}{qB_0} \right) = \frac{2v_0}{\alpha B_0}$$

$$\text{Now displacement} = \sqrt{(\text{Pitch})^2 + (\text{Diameter})^2}$$

$$= \frac{v_0}{\alpha B_0} \sqrt{4 + 9\pi^2}$$

4. Ans (B,C,D)

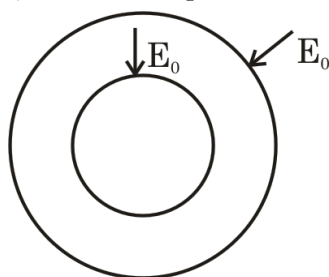
$$J = \sigma E = \frac{1}{\rho} E$$

$$J = (a + bh^2)E_0$$

Hence option (A) is incorrect. Current density increases with increase in height.

As field is directed downward electrons will flow upward (radially outward).

Now, to calculate charge density (ρ) we use gauss law (Assume a thin spherical shell as gaussian surface)

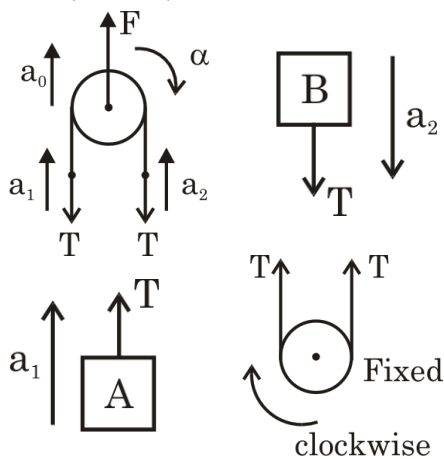


$$\phi = \phi_{IN} + \phi_{out} = \frac{q}{\epsilon_0}$$

$$\Rightarrow -E_0(4\pi r^2) + E_0 4\pi (r-dr)^2 = \frac{\rho (4\pi r^2 dr)}{\epsilon_0}$$

$$\Rightarrow \rho \approx -\frac{2\epsilon_0 E_0}{r} = -\frac{2\epsilon_0 E_0}{R+h}$$

5. Ans (A,C,D)



$$F = 2T \Rightarrow T = \frac{F}{2}$$

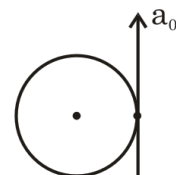
$$\text{acceleration of blocks, } a_1 = \frac{T}{M} = \frac{F}{2M} = a_2$$

both blocks have same acceleration but direction is opposite.

Net torque on fixed disc is zero. Torque due to tensions in the string cancel.

Acceleration of left pulley

$$a_0 = \frac{a_1 + a_2}{2} = \frac{F}{2M}$$



Left pulley

As the rope does not slip over pulley

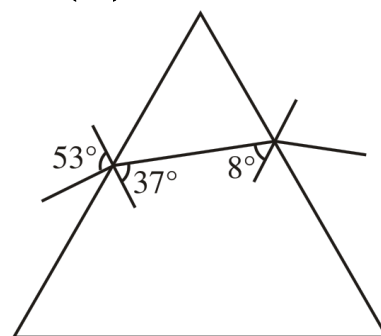
$$a_0 - \alpha R = a_2$$

$$\Rightarrow \frac{F}{2M} - \alpha R = \frac{R}{2M} \Rightarrow \alpha = 0$$

Hence left pulley does not rotate

Right pulley rotates in clockwise direction.

6. Ans (A)



Angle of incidence is the brewster's angle

$$\therefore \tan i = \mu = \frac{4}{3} \Rightarrow i = 53^\circ$$

Snell's law at the first interface.

$$\Rightarrow 1 \times \sin 53^\circ = \frac{4}{3} \sin r_1 \Rightarrow r_1 = 37^\circ$$

using relation $r_1 + r_2 = A$

$$\Rightarrow r_2 = 45^\circ - 37^\circ = 8^\circ$$

Snell's law at the second interface

$$\frac{4}{3} \sin 8^\circ = 1 \times \sin e$$

$$\Rightarrow \frac{4}{3} \left(8^\circ \times \frac{\pi}{180} \right) = e \times \frac{\pi}{180} \{ \because \sin \theta \approx \theta \}$$

$$\Rightarrow e = \frac{32}{3} \text{ degree}$$

Using $i + e = \delta + A$

$$\Rightarrow \delta = 53 + \frac{32}{3} - 45 = 18.7^\circ$$

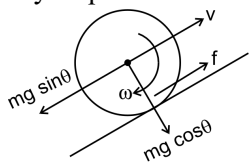
Reflected ray gets polarized. Transmitted ray remains as it is in terms of polarization. For no partial reflection light must be incident at brewster's angle and incident light must be polarized in a particular direction.

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

(A) For pure rolling motion $v = \omega R$, $mg \sin \theta$ will reduce v so ω must reduce, $mg \sin \theta$ & $mg \cos \theta$ will not produce any torque. So friction should be up the incline.

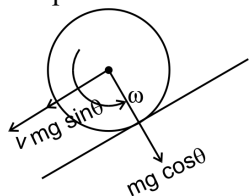


Now, Torque about centre $\neq 0$

Torque about point of contact $\neq 0$

A $\rightarrow$ P

(B) Again for pure rolling $v = \omega R$ should be valid, v will $\uparrow$ with time so ω should also $\uparrow$. So friction should be up the incline.

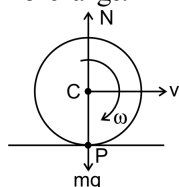


Torque about centre $\neq 0$

Torque about point of contact $\neq 0$

B $\rightarrow$ P

(C) Velocity v will not change with time so ω should not change.



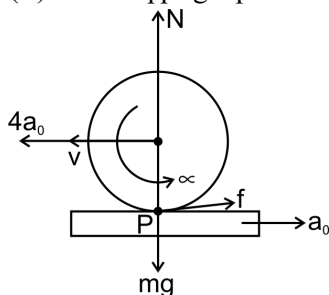
So friction should not be present $\tau_C = 0$

$\tau_C = 0 \Rightarrow$ Angular momentum is conserved about C.

$\tau_P = 0 \Rightarrow$ Angular momentum is conserved about P.

C $\rightarrow$ S, R, T

(D) For no slipping at point of contact



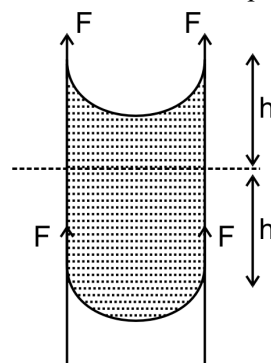
$$\alpha R - 4a_0 = a_0 \Rightarrow \alpha = \frac{5a_0}{R}$$

For this angular acc. Torque should be present about centre. So friction should be in opposite direction of velocity. Torque about P is zero $\Rightarrow$ Angular momentum is conserved about point P.

D $\rightarrow$ Q, R

8. **Ans (C)**

(A) As liquid rises upto height h due to surface tension which means force of surface tension can lift or handle liquid upto height h

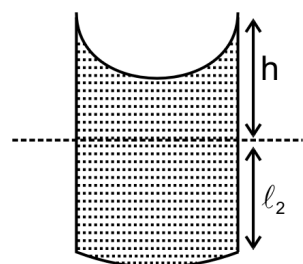


So, Height of liquid is $2h$

& Both surfaces are hemispherical.

A $\rightarrow$ P, Q

(B) As $\ell_1 > h \Rightarrow$ Length of liquid above liquid surface is h



But $\ell_2 < h \Rightarrow$ Length of liquid in capillary below liquid surface is ℓ_2

Overall length in liquid in capillary $= \ell_2 + h$

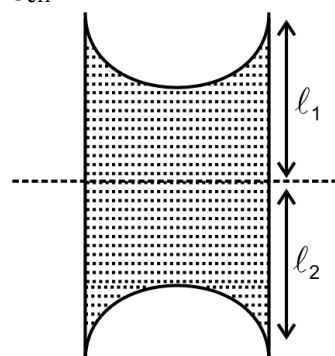
Upper surface $\rightarrow$ Hemispherical

Lower surface $\rightarrow$ A little flat with respect to hemispherical surface

B $\rightarrow$ T

(C) In absence of gravity liquid will fill entire length of capillary

$$g_{\text{eff}} = 0 \Rightarrow h \rightarrow \infty$$



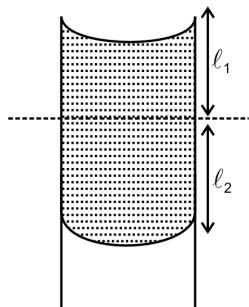
As atmospheric pressure is present therefore when we observe liquid surface from top it should appear concave.

$C \rightarrow P, R$

(D) $\ell_1 < h \Rightarrow$ Capillary is of insufficient length so liquid will rise upto length ℓ_1 but surface is not fully hemispherical.

As $\ell_2 < h \Rightarrow$ Liquid will be in all length but surface is not fully hemispherical.

$C \rightarrow R, T$



9. **Ans (B)**

$$R_1 = 2\rho \cdot \frac{2\ell}{\pi r^2} = 4\rho \cdot \frac{\ell}{A}$$

$$R_2 = \rho \cdot \frac{2\ell}{\pi(3r)^2} = \frac{2}{9}\rho \cdot \frac{\ell}{A}$$

$$R_3 = 3\rho \cdot \frac{3\ell}{\pi(2r)^2} = \frac{9}{4}\rho \cdot \frac{\ell}{A}$$

$$\frac{P_2 P_3}{P_1^2} = \frac{R_2 R_3}{(R_1)^3} = \frac{\frac{2}{9} \times \frac{9}{4}}{16} = \frac{1}{32}$$

$$E = \rho J$$

$$J = nev_d = \frac{I}{A}$$

$$E = \frac{\rho \cdot I}{A}$$

$$E_1 = \frac{2\rho \cdot I}{\pi r^2} = \frac{2\rho I}{A}$$

$$E_2 = \frac{\rho \cdot I}{\pi(3r)^2} = \frac{\rho I}{9A}$$

$$E_3 = \frac{3\rho \cdot I}{\pi(2r)^2} = \frac{3\rho I}{4A}$$

$$\frac{E_1 E_2}{E_3^2} = \frac{2 \times \frac{1}{9}}{\frac{9}{16}} = \frac{32}{81}$$

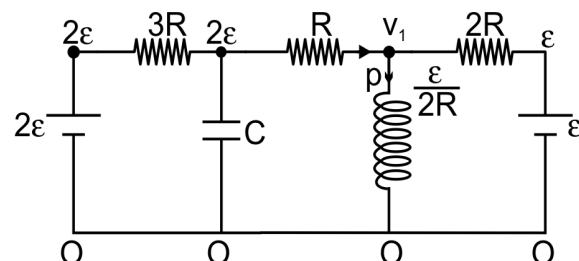
Similarly value of

$$\frac{v_{d2} v_{d3}}{v_{d1}^2} = \frac{1}{4}$$

$$\frac{V_2 \cdot V_3}{V_1^2} = \frac{1}{32}$$

10. **Ans (A)**

Just after S_3 and S_4 are closed current in inductor is unchanged and charge on capacitor is also unchanged.



KCL at junction P

$$\frac{v_1 - 2\epsilon}{R} + \frac{v_1 - 2\epsilon}{2R} + \frac{\epsilon}{2R} = 0$$

$$v_1 = \frac{4\epsilon}{3}$$

So, current in resistance R is

$$I_1 = \frac{2\epsilon - \frac{4\epsilon}{3}}{R} = \frac{2\epsilon}{3R} = 2.5A$$

$$\text{Similarly, } I_2 = -\frac{2\epsilon}{3R} = -2.5A$$

Current through capacitor = 2.5A

$$\text{So, } u = \frac{Q^2}{2C}$$

$$\begin{aligned} \frac{dU}{dt} &= \frac{1}{2C} \cdot 2Q \cdot \frac{dQ}{dt} \\ &= \frac{1}{2C} \cdot 2 \cdot C v \cdot \frac{dQ}{dt} \\ &= v \cdot \frac{dQ}{dt} = 2\epsilon(2.5) = 20 \times 2.5 = 50 \end{aligned}$$

As, $\frac{dQ}{dt}$ is -ve So, $\frac{du}{dt}$ is also negative

$$\text{In inductor, } L \frac{di}{dt} = \frac{4\epsilon}{3}; \frac{di}{dt} = \frac{4\epsilon}{3}$$

$$B = \mu_0 n i$$

$$\frac{dB}{dt} = \mu_0 n \frac{4\epsilon}{3L}$$

$$L = \mu_0 n^2 \pi r^2 \ell$$

$$\begin{aligned} \frac{dB}{dt} &= \mu_0 n \frac{4\epsilon}{3 \cdot \mu_0 \cdot n^2 \pi r^2 \ell} \\ \left(\because \pi r^2 \ell &= \frac{16}{3} \times 10^{-3} \right) \end{aligned}$$

$$\frac{dB}{dt} = \frac{dB}{dt} = \frac{4}{3} \times \frac{10}{16} \times 3 = 2.5$$

As, $\frac{di}{dt}$ is positive, So, $\frac{dB}{dt}$ is also positive.

PART-1 : PHYSICS

SECTION-II

1. **Ans (4.00)**

$$V_0 = IR + \frac{I^2}{K}$$

$$\Rightarrow I^2 + (RK)I - V_0K = 0$$

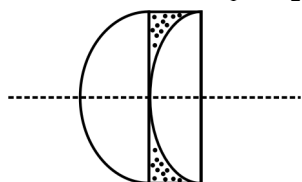
$$I = \frac{-RK + \sqrt{R^2K^2 + 4V_0K}}{2}$$

2. **Ans (2.00)**

Focal length of Equiconvex lens

$$\Rightarrow \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For Equiconvex $R_1 = R_2 = R$



$$\Rightarrow \frac{1}{20} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$\Rightarrow \frac{1}{20} = \frac{1}{2} \cdot \frac{2}{R} \Rightarrow \boxed{R = 20\text{cm}}$$

Radius of curvature of lens is 20 cm

If light rays parallel to principle axis are not deviated then image is formed at infinite.

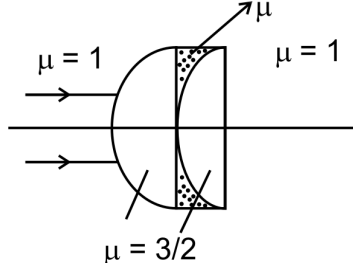
Consider RI of liquid is μ .

Applying formula for image formation

$$\frac{1}{v} - \frac{1}{u} = \frac{\frac{3}{2} - 1}{20} + \frac{\mu - \frac{3}{2}}{\infty} + \frac{\frac{3}{2} - \mu}{20} + \frac{1 - \frac{3}{2}}{\infty}$$

As $u \rightarrow \infty, v \rightarrow \infty$

$$\frac{1}{\infty} - \frac{1}{\infty} = \frac{\frac{1}{2}}{20} + 0 + \frac{\frac{3}{2} - \mu}{20} + 0$$



$$0 = \frac{1}{40} + \frac{\frac{3}{2} - \mu}{20}$$

$$\Rightarrow -\frac{1}{2} = \frac{3}{2} - \mu$$

$$\Rightarrow \mu = \frac{3}{2} + \frac{1}{2} = \frac{4}{2} = 2$$

3. **Ans (0.31)**

$$\text{Energy in the light } E = \frac{hc}{\lambda} = \frac{12400\text{eV}\text{\AA}}{4000\text{\AA}} = 3.1\text{eV}$$

10% of this energy is lost by electron in 1st collision

So energy remaining after a 1st collision

$$= 3.1 - .31\text{eV} = 2.79\text{eV}$$

Again 10% energy is lost in 2nd collision.

Energy remaining after 2nd collision

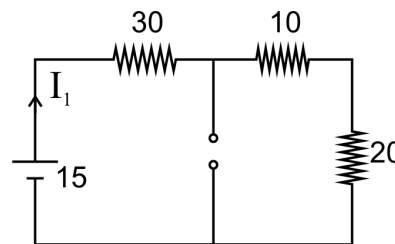
$$= 2.79\text{eV} - .279\text{eV} = 2.511$$

Now KE of electron after being emitted by surface

$$= 2.511 - 2.2\text{eV} = 0.31\text{eV}$$

4. **Ans (0.25)**

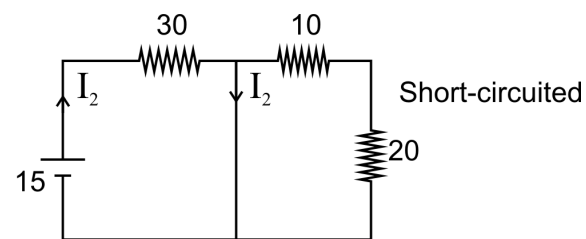
At $t \rightarrow 0, R_L \rightarrow \infty$, so circuit can be reduced to



According to graph current at this time is I_1

$$\Rightarrow I_1 = \frac{15}{30 + 10 + 20} = \frac{15}{60} = \frac{1}{4} = 0.25\text{Amp}$$

At $t \rightarrow \infty, R_L \rightarrow 0$ Now circuit can be reduced to



$$I_2 = \frac{15}{30} = \frac{1}{2}$$

$$\Rightarrow I_2 - I_1 = \frac{1}{2} - \frac{1}{4} = 0.5 - 0.25 = 0.25$$

5. Ans (7.00)

$$U = 5 - 20x + 2x^2$$

$$\text{Force on particle } F = -\frac{dU}{dx} = -(0 - 20 + 4x) = 20 - 4x$$

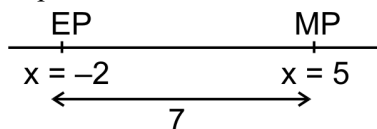
$$F = 20 - 4x \Rightarrow \text{acceleration } a = \frac{F}{m} = \frac{-4x + 20}{2}$$

$$\Rightarrow [a = -2x + 10] \text{ \{Equation of SHM\}}$$

$$\text{At mean position } F = 0 \Rightarrow [x = 5]$$

Therefore $x = 5$ is mean position of particle.

As particle is released at $x = -2$ (Extreme position) therefore amplitude is 7



$$\text{Now } a = -2x + 10$$

$$\Rightarrow \omega^2 = 2$$

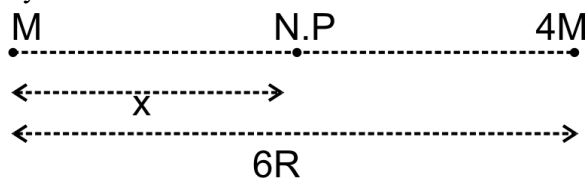
$$\Rightarrow \omega = \sqrt{2}$$

$$\text{And } V_{\max} = A\omega = 7\sqrt{2}$$

$$\Rightarrow k = 7$$

6. Ans (3.00)

Consider mass of the particle is m . We have to give speed to particle so that it can reach to neutral point of system. As particle crosses NP it will be pulled by $4M$.

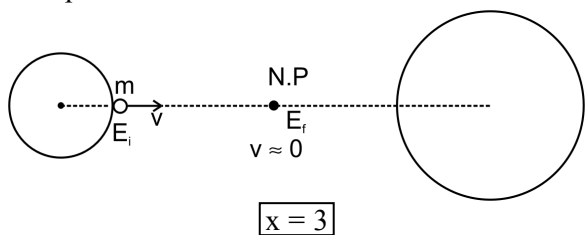


At N.P, gravitational field is zero.

$$\text{therefore, } \frac{GM}{x^2} = \frac{G \cdot 4M}{(6R - x)^2}$$

$$\Rightarrow \frac{1}{x} = \frac{2}{6R - x} \Rightarrow [x = 2R]$$

Now applying energy conservation for initial and final positions.



$$E_i = E_f$$

$$\Rightarrow \frac{1}{2}mv^2 + \left(-\frac{GMm}{R} - \frac{G4Mm}{5R} \right)$$

$$= \frac{1}{2}m(0)^2 + \left(-\frac{GMm}{2R} - \frac{Gm4M}{4R} \right)$$

$$\Rightarrow \frac{v^2}{2} + \left(-\frac{9}{5} \frac{GM}{R} \right) = -\frac{GM}{R} \left(\frac{1}{2} + 1 \right)$$

$$\Rightarrow \frac{v^2}{2} = \frac{9}{5} \frac{GM}{R} - \frac{GM}{R} \cdot \frac{3}{2} = \frac{GM}{R} \left(\frac{9}{5} - \frac{3}{2} \right)$$

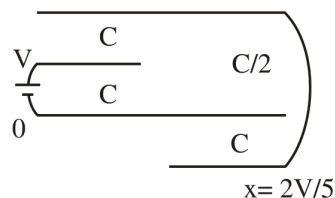
$$\Rightarrow \frac{v^2}{2} = \frac{3}{10} \frac{GM}{R} \Rightarrow v^2 = \frac{3}{5} \frac{GM}{R}$$

$$\Rightarrow v = \sqrt{\frac{3}{5} \frac{GM}{R}} \Rightarrow [x = 3]$$

7. Ans (1.60)

$$x = 2V/5$$

$$x = 2V/5$$



$$C(x - v) + \frac{\ell}{2}(x - 0) + C(x - 0) = 0$$

$$\frac{5x}{2} = V$$

$$x = \frac{2V}{5}$$

Q on plate connected to +ve terminal =

$$C \left(V - \frac{2V}{5} \right) + C(V - 0)$$

$$= \frac{8CV}{5} = C_{eq} V$$

$$\Rightarrow C_{eq} = \frac{1.6\epsilon_0 L a}{d}$$

8. Ans (5.00)

Path difference produced by glass slab =

$$\left(1 - \frac{\mu_2}{\mu_1} \right) t$$

$\mu_2 \rightarrow$ R.I. of film

$\mu_1 \rightarrow$ R.I. of medium

$$\text{Now, } \Delta x = \left(1 - \frac{\mu_2}{\mu_1} \right) t = \frac{n \cdot \lambda}{\mu}$$

$$\Rightarrow \left(1 - \frac{1}{3/2} \right) t = \frac{n \cdot \lambda}{\mu}$$

$$\Rightarrow \left(1 - \frac{2}{3} \right) t = \frac{n \cdot \lambda \cdot 2}{3} \Rightarrow \frac{t}{3} = \frac{2n\lambda}{3}$$

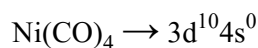
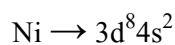
$$\Rightarrow n = \frac{t}{2\lambda} = \frac{4 \times 10^{-6}}{2 \times 4000 \times 10^{-7}}$$

$$= \frac{4 \times 10}{2 \times 40} = \frac{40}{8} = 5$$

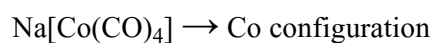
PART-2 : CHEMISTRY

SECTION-I (i)

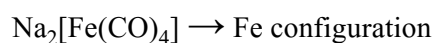
1. **Ans (D)**



→ sp^3 hybridization tetrahedral shape

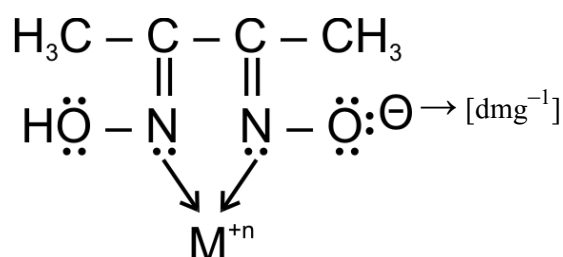
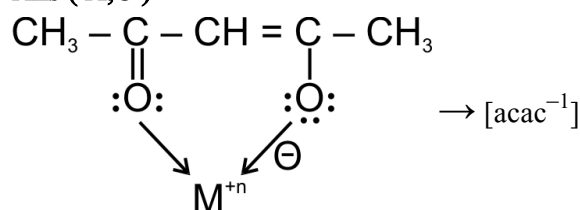


→ $3d^{10}4s^0$ Tetrahedral shape sp^3 hybridization

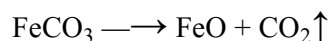
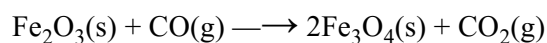
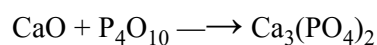
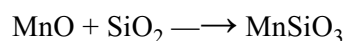


→ $3d^{10}4s^0$ Tetrahedral shape sp^3 hybridization

2. Ans (A,C)



4. **Ans (A,B,C,D)**



All reactions occur during extraction of iron wrought form haematite.

5. Ans (A,B,D)

For a polytropic process

$$pv^x = \text{const} = k$$

$$\frac{dP}{dV} = -x \frac{P}{V}$$

For 2 moles of the ideal gas

$$pv = 2RT$$

$$V = \frac{2RT}{P}$$

$$\frac{dP}{dV} = -x \frac{P^2}{2RT}$$

$$\frac{dP}{dV} = - \frac{P^2}{2RT} = \frac{-xP^2}{2RT}$$

by comparing we will get $x = -1$

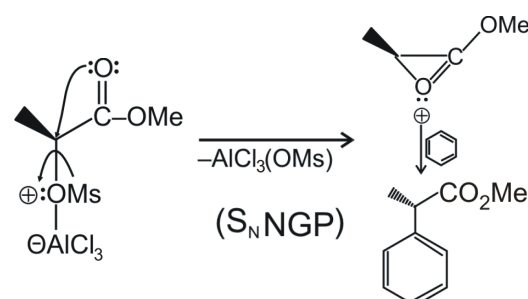
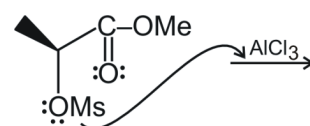
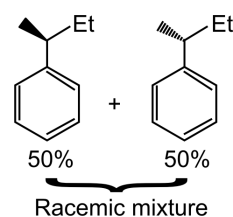
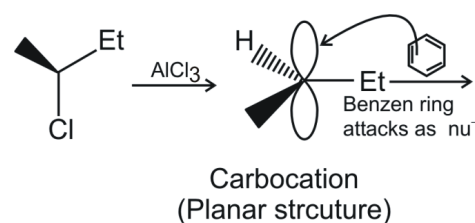
$$C_M = C_{v,m} + \frac{R}{1-x} = \frac{5}{2}R + \frac{R}{1-(-1)} = 3R$$

$$C_m = \frac{5}{2}R + \frac{R}{2} = 3R$$

$$\mathrm{TV}^{-1-1} = \mathbf{K}$$

$$T = KV^2$$

6. **Ans (B)**



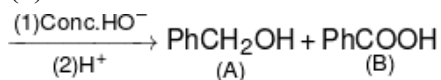
Retention of the configuration.

PART-2 : CHEMISTRY

SECTION-I (ii)

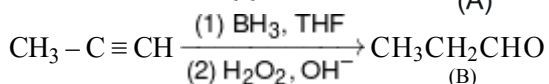
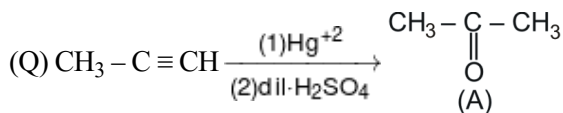
7. Ans (A)

(P) Ph - CHO



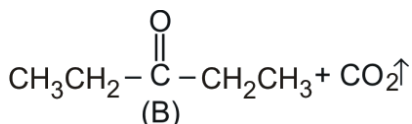
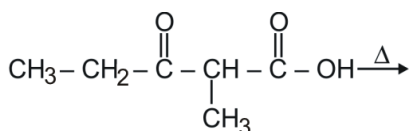
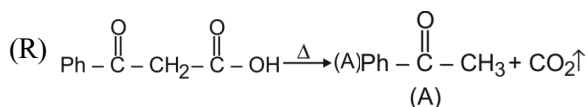
→ Can be differentiated by $NaHCO_3$

P → 3



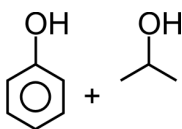
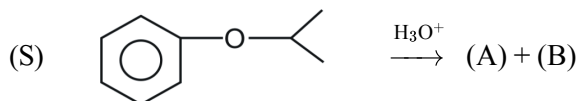
(A) and (B) can be differentiated by Tollen's test, fehling solution & iodoform test

Q → 1, 2, 4



(A) & (B) can differentiated only iodoform test

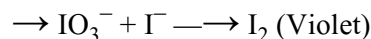
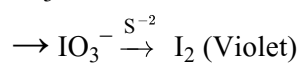
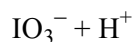
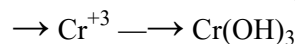
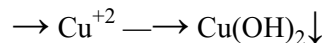
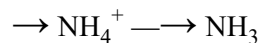
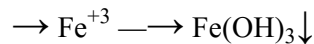
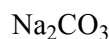
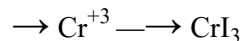
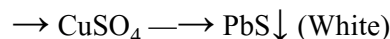
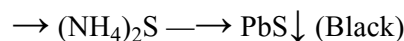
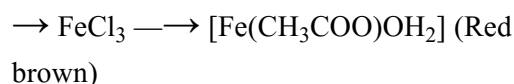
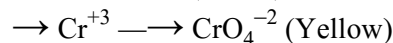
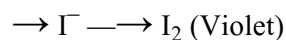
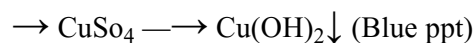
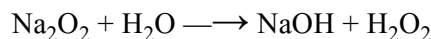
R → 4



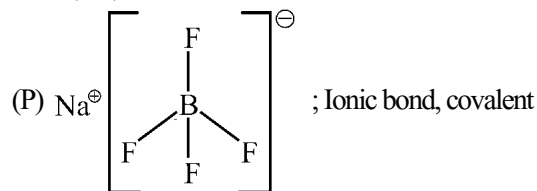
(A) & (B) can be differentiated by iodoform test & neutral $FeCl_3$

S → 4, 5

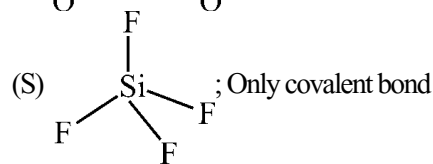
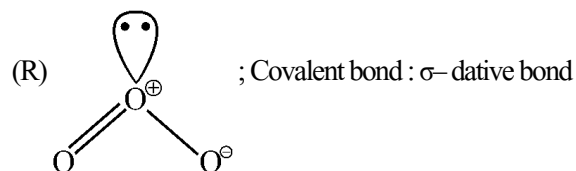
8. Ans (B)



10. Ans (A)



bond and σ -dative bond

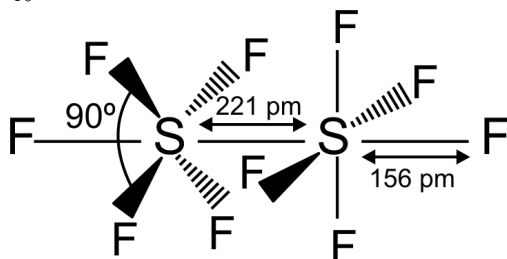


[All the above molecule stricly follow octet rule]

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.00)**



Maximum number of atoms lie in same plane = 6

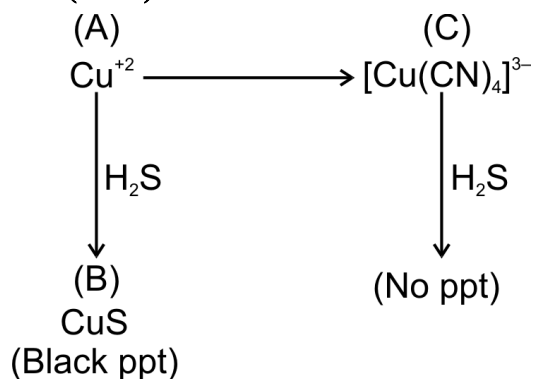
2. **Ans (250.00)**

Millimol of $H^+ = 0.005 = m$ mol of OH^-

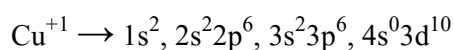
millimol of $SO_4^{2-} = 0.005$

$$\text{ppm (hardness)} = \frac{0.005 \times 100}{2000} \times 10^6 = 250$$

3. **Ans (0.00)**

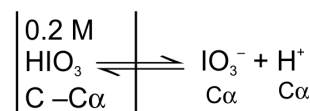
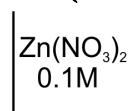


Complex (C) $\Rightarrow [Cu(CN)_4]^{3-}$

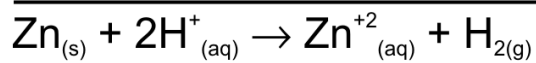
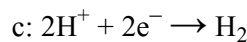
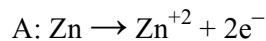


$\therefore$ Number of unpaired $e^- = 0$

4. **Ans (5.00)**



$$E_{\text{cell}}^{\circ} = 0.76V$$



$$K_a = \frac{C\alpha^2}{1 - \alpha}$$

$$\Rightarrow 1 \times 10^{-1} = 2 \times 10^{-1} \times \frac{\alpha^2}{1 - \alpha}$$

$$\Rightarrow 1 - \alpha = 2\alpha^2$$

$$\Rightarrow 2\alpha^2 + \alpha - 1 = 0$$

$$\Rightarrow \alpha = -1 \pm \frac{\sqrt{1 + 4 \times 2}}{4}$$

$$\Rightarrow \alpha = \frac{-1 \pm 3}{4} = 0.5$$

$$\Rightarrow (H^+) = 0.1 M$$

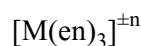
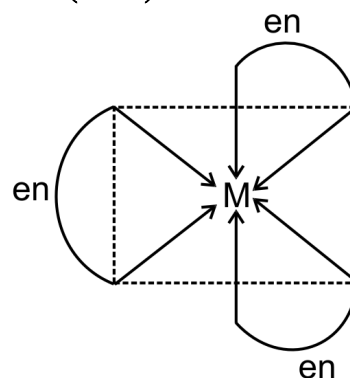
$$\text{Now, } E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log \frac{(Zn^{+2})(P_{H_2})}{(H^+)^2}$$

$$= 0.76 - \frac{0.0591}{2} \log \frac{10^{-1} \times 10^{-1}}{(10^{-1})^2}$$

$$= 0.76 - \frac{0.0591}{2} \log(1)$$

$$= 0.76 V = X$$

5. **Ans (5.00)**



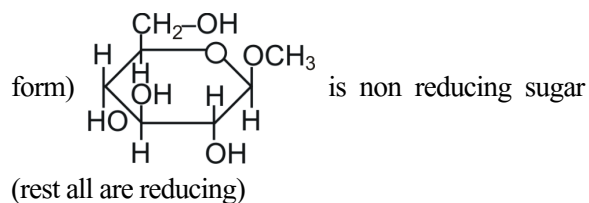
Bidentate ligand Denticity = 2

Number of ligands = 3

$$P + Q = 5$$

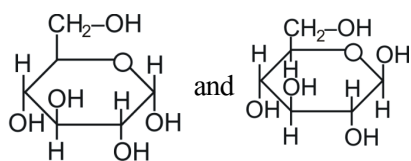
6. Ans (5.00)

Reducing sugars are those sugars in which anomeric carbon atom has – OH group (Hemiacetal form)

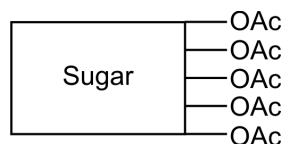
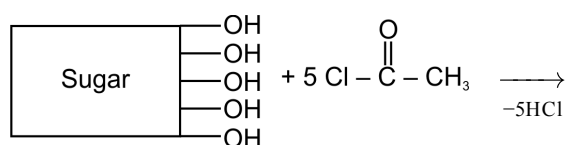


$$x = 7$$

→ Only these two sugars will be form pentactate because five –OH groups are present



$$y = 2$$

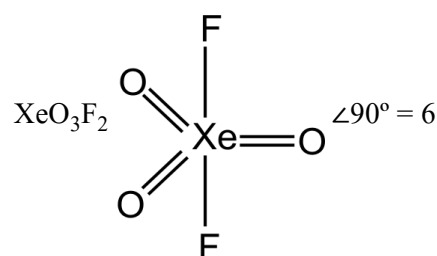
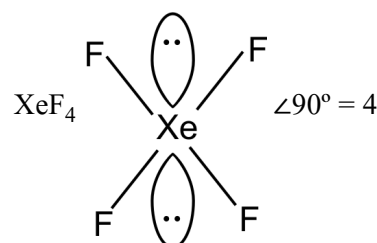
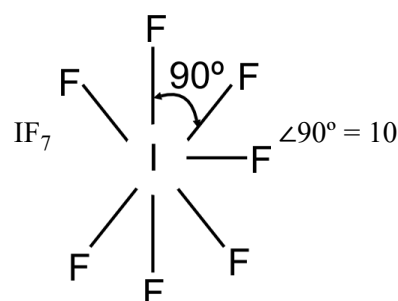
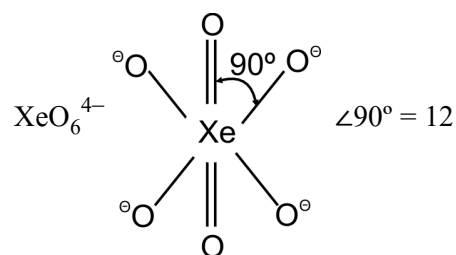
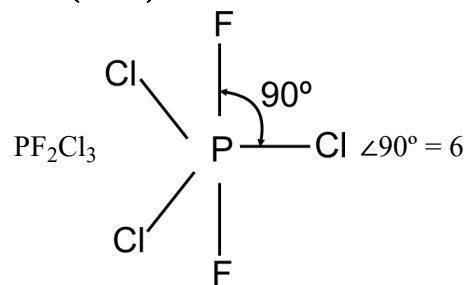


Addition of one $-\text{C}(=\text{O})-\text{CH}_3$ group will change in mol. wt. by 42 units

$$\begin{aligned}
 \therefore \text{Number of the } \text{CH}_3-\text{C}(=\text{O})-\text{groups added} \\
 &= \frac{210}{42} = 5
 \end{aligned}$$

$$x - y = 7 - 2 = 5$$

7. Ans (5.00)



PF_2Cl_3 , XeO_6^{4-} , IF_7 , XeF_4 , XeO_3F_2 have at least one bond angle of $90^\circ = 5$

8. **Ans (5.00)**

Amount of Ag deposited = $(675 \times 0.0004) \times 10$ gram

$$\frac{w}{E} = \frac{I \times t}{96500} \text{ (Faraday's 2nd law)}$$

$$\frac{(675 \times 0.0004) \times 10}{108} = \frac{96.5 \times t}{2 \times 96500} \text{ [Eq. wt of Ag]}$$

$t = 50$ sec.

$$\text{Ans} = \frac{x}{10}$$

$$\Rightarrow \frac{50}{10} = 5$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C,D)**

$$(a-1)(x^2 + \sqrt{3}x + 1)^2 - (a+1)[(x^2+1)^2 - (x\sqrt{3})^2] \leq 0$$

$$\text{or } (a-1)(x^2 + \sqrt{3}x + 1)^2 - (a+1)[x^2 + x\sqrt{3} + 1]$$

$$(x^2 - x\sqrt{3} + 1) \leq 0$$

$$(x^2 - x\sqrt{3} + 1)[(a-1)(x^2 + \sqrt{3}x + 1) - (a+1)]$$

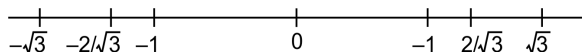
$$(x^2 - x\sqrt{3} + 1) \leq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow -2(x^2 + 1) + 2a\sqrt{3}x \leq 0$$

$$\Rightarrow x^2 - a\sqrt{3}x + 1 \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow 3a^2 - 4 \leq 0 \text{ (D} \leq 0)$$

$$\Rightarrow a \in \left[-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}} \right]$$



$\Rightarrow$ Number of possible integral value of a is $\{-1, 0, 1\}$

$\Rightarrow 3$

and sum of all integral values of a is $-1 + 0 + 1 = 0$

2. **Ans (B,C,D)**

$$f(x) = \int_0^1 5 + (1-t)dt + \int_1^x 5 - (1-t)dt$$

$$\Rightarrow f(x) = \begin{cases} 5x + 1, & x \leq 2 \\ \frac{x^2}{2} + 4x + 1, & x > 2 \end{cases}$$

3. **Ans (A,C)**

$$\Delta = 0, a = 3$$

$$\int_0^{-6} f(x)dx = \int_0^{-3} f(x)dx + \int_{-3}^{-6} f(x)dx$$

$$= \int_{-3}^{-6} f(x+3)dx + \int_{-3}^{-6} f(x)dx$$

$$= 4 \int_{-3}^{-6} dx = -12$$

4. **Ans (B,C,D)**

$$\int \frac{(\sec^2 x - 1) \tan x}{\tan^2 x + 2} dx$$

$$= \int \left(\frac{\sec^2 x \tan x}{\sec^2 x + 1} - \frac{\tan x}{\sec^2 x + 1} \right) dx$$

Put $\sec x = t \Rightarrow \sec x \tan x dx = dt$

$$\int \left(\frac{t}{t^2 + 1} - \frac{1}{t(t^2 + 1)} \right) dt$$

$$= \int \frac{t^2 - 1}{t(t^2 + 1)} dt = \int \frac{1 - \frac{1}{t^2}}{t + \frac{1}{t}} dt$$

$$= \ln \left| \left(t + \frac{1}{t} \right) \right| + C$$

$$= \ln \left(\frac{1 + \cos^2 x}{\cos x} \right) + C$$

$$g(x) = \cos^2 x$$

Now, verify the options.

5. **Ans (A,B,C)**

$a_{n/2}$ and $a_{(n+1)/2}$ are both 1. Note that the number of ways to choose numbers left of $a_{n/2}$ and $a_{\frac{n+1}{2}}$

$\left(a_1 > a_2 > a_3 \dots a_{\frac{n}{2}}; a_{\frac{n+1}{2}} \right)$ is same as that of right. We started with 9

$$n = 9 \Rightarrow {}^8C_4 = 70 \text{ (rejected)}$$

$$n = 10 \Rightarrow {}^9C_4 = 126$$

$$n = 11 \Rightarrow {}^{10}C_5 = 252$$

$$n = 12 \Rightarrow {}^{11}C_5 = 462$$

$$n = 13 \Rightarrow {}^{12}C_6 = 924 \text{ (rejected)}$$

6. **Ans (A,B)**

Given $a \neq b \neq c$, $a, b, c \in \mathbb{R}$

Now $ax^2 + bx + c \geq 0 \Rightarrow b^2 - 4ac \leq 0$ and $a > 0$ (1)

$bx^2 + cx + a \geq 0 \Rightarrow c^2 - 4ab \leq 0$ and $b > 0$ (2)

$cx^2 + ax + b \geq 0 \Rightarrow a^2 - 4bc \leq 0$ and $c > 0$ (3)

Equality can not hold simultaneously ($\because a \neq b \neq c$)

$$\therefore \frac{a^2 + b^2 + c^2}{ab + bc + ca} < 4$$

and since $a \neq b \neq c$

$$\Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 > 0$$

$$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} > 1$$

$$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} \in (1, 4)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (D)**

(P) Let $g(x) = f(x) e^{\cos x}$

$$\Rightarrow g'(x) = f'(x) e^{\cos x} - f(x) \sin x e^{\cos x} = e^{\cos x} (f'(x) - f(x) \sin x) \leq 0$$

$\because g(x)$ is a decreasing function

$$\therefore g(x) \leq g(0) = 0 \text{ i.e. } f(x) e^{\cos x} \leq 0 \text{ but } f(x) \geq 0$$

$$\text{and } e^{\cos x} > 0 \Rightarrow f(x) = 0$$

$$\therefore f\left(\frac{3\pi}{2}\right) = 0.$$

$$(Q) (a-3)x^2 + 12x + 6 + a - (\sin^{-1} \sin 100 + \cos^{-1} \cos 100) < 0 \Rightarrow a < 3, D < 0$$

$$144 - 4(a+6)(a-3) < 0 \Rightarrow a < -9 \text{ or } a > 6,$$

$$(R) P\left(\frac{B}{A \cup B^c}\right) = \frac{P(A \cap B)}{P(A \cup B^c)} = \frac{1}{5}$$

$$(S) \cos^{-1}(4x^3 - 3x) - \frac{\pi}{2} \geq 0 \Rightarrow -1 \leq 4x^3 - 3x \leq 0 \text{ i.e.}$$

$$x \in \left[-4, -\frac{\sqrt{3}}{2}\right] \cup \left[0, \frac{\sqrt{3}}{2}\right]$$

$$\text{also } \frac{(2x+3)!}{\sqrt{x+2}} \text{ is defined } \forall x > -2 \text{ and } (2x+3) \in \mathbb{W}$$

$$x = -1, 0, \frac{1}{2}.$$

8. **Ans (B)**

$$3x + y - z = 0 \quad \dots(1)$$

$$x - \frac{py}{4} + z = 0 \quad \dots(2)$$

$$2x - y + 2z = q \quad \dots(3)$$

$$(2) \times 2 - (3)$$

$$\Rightarrow \left(1 - \frac{p}{2}\right)y = 4 - q$$

For unique solution, $p \neq 2$, $q \in \mathbb{N} \Rightarrow$ Number of

ordered pairs (p, q) in $[1, 10]$ are 90. For

infinite solution, $p = 2$ and $q = 4$

$\Rightarrow$ exactly one ordered pairs

For no solution, $p = 2$ and $q \neq 4 \Rightarrow$ Number of

ordered pairs (p, q) in $[1, 10]$ are 9.

9. **Ans (D)**

Let E be the event of getting 2 head and 1 tail

$$P(A) = P(B) = P(C) = \frac{1}{3}$$

$$\text{Using Bayes' theorem } P(A/E) = \frac{9}{25},$$

$$P(B/E) = \frac{32}{75}$$

10. Ans (C)

$$R = P^T Q^K P = P^T (PAP^T)^K P = P^T PAP^T PAP^T \dots\dots\dots$$

$$PAP^T P \text{ (n times)} = A^K \text{ as } PP^T = 1 \text{ as } P \text{ is}$$

orthogonal

$$A^K = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$$

$$T = P^T S^K P = B^K$$

$$B^K = \begin{bmatrix} a^k & \frac{6(a^k-1)}{a-1} \\ 0 & 1 \end{bmatrix}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$1 = \int_{1/e}^{\cos x} \frac{t}{1+t^2} dt + \int_{1/e}^{\sec x} \frac{dt}{t(1+t^2)}$$

$$\text{Put } t = \frac{1}{z} \Rightarrow dt = -\frac{1}{z^2} dz$$

$$I = \int_{1/e}^{\cos x} \frac{t}{1+t^2} dt + \int_e^{\cos x} \frac{-\frac{1}{z^2}}{\frac{1}{z} \left(1 + \frac{1}{z^2}\right)} dz$$

$$\Rightarrow I = \int_{1/e}^{\cos x} \frac{t}{1+t^2} dt - \int_e^{\cos x} \frac{z}{z^2+1} dz$$

$$\Rightarrow I = \int_{1/e}^{\cos x} \frac{t}{1+t^2} dt + \int_{\cos x}^e \frac{t}{1+t^2} dt$$

$$\Rightarrow I = \int_{1/e}^e \frac{t}{1+t^2} dt = \frac{1}{2} \ln(1+t^2) \Big|_{1/e}^e$$

$$= \frac{1}{2} \left[\ln(1+e^2) - \ln 1 + \frac{1}{e^2} \right]$$

$$\Rightarrow I = \frac{1}{2} [2] = 1$$

2. Ans (5.00)

$$\text{Let } \sum_{i=1}^6 x_i = \alpha \text{ and } \sum_{i=1}^6 y_i = \beta$$

Let O be the orthocentre of the triangle made by

$(x_1, y_1), (x_2, y_2)$ and (x_3, y_3)

$$\Rightarrow O \text{ is } (x_1 + x_2 + x_3, y_1 + y_2 + y_3) \equiv (\alpha_1, \beta_1)$$

Similarly let G be the centroid of the triangle made

by other three points

$$\Rightarrow G \text{ is } \left(\frac{x_4 + x_5 + x_6}{3}, \frac{y_4 + y_5 + y_6}{3} \right) \Rightarrow G \text{ is } \left(\frac{\alpha - \alpha_1}{3}, \frac{\beta - \beta_1}{3} \right)$$

The point dividing OG in the ratio 3 : 1 is

$$\left(\frac{\alpha}{4}, \frac{\beta}{4} \right) \equiv (3, 2) \Rightarrow h + k = 5$$

3. Ans (1356.00)

$$/ \boxed{\text{LOVE}} / S / L / F / E / \Rightarrow {}^6C_2 \times (1)^2 \times 5!$$

(Fill I in the gaps)

Now

$$/ \boxed{\text{LLOVE}} / S / F / E / \Rightarrow {}^5C_2 \times (1)^2 \times 4!$$

Same for

$$/ \boxed{\text{LOVEE}} / S / F / L / \Rightarrow {}^5C_2 \times (1)^2 \times 4!$$

$$/ \boxed{\text{LLOVEE}} / S / F / \Rightarrow {}^4C_2 \times (1)^2 \times 3!$$

$$\text{So Ans.} = {}^6C_2 \times (1)^2 \times 5! - {}^5C_2 \times (1)^2 \times 4! - {}^5C_2 \times$$

$$(1)^2 \times 4! + {}^4C_2 \times (1)^2 \times 3!$$

$$= 1356$$

4. Ans (4.00)

$$\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$$

any point P on the curve $(4(1 + \cos\theta), 3(1 + \sin\theta))$.

The reflection of the point P about the line $x - y - 2 = 0$ can be found using

$$\frac{h - 4(1 + \cos\theta)}{1} = \frac{k - 3(1 + \sin\theta)}{-1}$$

$$= \frac{-2(4(1 + \cos\theta) - 3(1 + \sin\theta) - 2)}{2}$$

$$= -(4\cos\theta - 3\sin\theta) + 1$$

$$h = 4(1 + \cos\theta) - 4\cos\theta + 3\sin\theta + 1 = 4 + 3\sin\theta + 1$$

$$\text{and } k = 3(1 + \sin\theta) + 4\cos\theta - 3\sin\theta = 3 + 4\cos\theta + 1$$

$$\left(\frac{h-5}{3}\right)^2 + \left(\frac{k-2}{4}\right)^2 = 1$$

$$\therefore \text{equation of image } 16(x-5)^2 + 9(y-2)^2 = 144$$

$$16x^2 + 9y^2 - 160x - 36y + 292 = 0$$

$$k_1 = 16, k_2 = 36$$

$$\Rightarrow k_1 + k_2 = 52$$

5. Ans (2.00)

Let $P(x_1, y_1, z_1)$ be a point on $ax + by + cz = d$.
Then

$$ax_1 + by_1 + cz_1 = d \quad \dots(i)$$

Let $OP = r$. Then direction cosines of OP are

$$\frac{x_1}{r}, \frac{y_1}{r}, \frac{z_1}{r}$$

$$\text{Equation of line OP is } \frac{x-0}{\frac{x_1}{r}} = \frac{y-0}{\frac{y_1}{r}} = \frac{z-0}{\frac{z_1}{r}} = \lambda$$

Let $Q(\alpha, \beta, \gamma)$ be a point on OP such then $OQ = \lambda$.

$$\text{Then coordinates of } Q\left(\frac{x_1\lambda}{r}, \frac{y_1\lambda}{r}, \frac{z_1\lambda}{r}\right).$$

But coordinate of Q be (α, β, γ)

$$x_1 = \frac{\alpha r}{\lambda}, y_1 = \frac{\beta r}{\lambda}, z_1 = \frac{\gamma r}{\lambda}$$

$$\Rightarrow ax_1 + by_1 + cz_1 = d$$

$$\frac{r}{\lambda}(a\alpha + b\beta + c\gamma) = d \quad \dots(ii)$$

$$OP \cdot OQ = d^2 \quad \dots(iii)$$

$$r \cdot \lambda = d^2$$

$$\frac{\lambda^2 d}{(a\alpha + b\beta + c\gamma)} = d^2$$

$$\lambda^2 = d(a\alpha + b\beta + c\gamma)$$

$$\alpha^2 + \beta^2 + \gamma^2 = d(a\alpha + b\beta + c\gamma)$$

6. Ans (5.00)

$$P\left(\frac{R}{A}\right) = \left(\frac{2}{3}\right)^n, P\left(\frac{R}{B}\right) = \left(\frac{1}{3}\right)^n, \text{ in } n \text{ throws}$$

$$P(R) = \frac{1}{2}\left(\frac{2}{3}\right)^n + \frac{1}{2}\left(\frac{1}{3}\right)^n, \text{ in } n \text{ throws}$$

$$P\left(\frac{A}{R}\right) = \frac{P\left(\frac{R}{A}\right)P(A)}{P(R)}$$

$$= \frac{\frac{1}{2}\left(\frac{2}{3}\right)^n}{\frac{1}{2}\left(\frac{2}{3}\right)^n + \frac{1}{2}\left(\frac{1}{3}\right)^n}$$

$$= \frac{2^n}{2^n + 1} = \frac{32}{33} \Rightarrow n = 5$$

7. Ans (5.00)

$$B^{-1} = \frac{\text{adj}B}{|B|}$$

$$= \frac{\text{adj}(\text{adj}A)}{|\text{adj}A|} = \frac{|A|^{n-2}A}{|A|^{n-1}}$$

$$= \frac{A}{|A|} = \frac{A}{3} \Rightarrow B^{-1} = \frac{A}{3}$$

$$f(x) = \frac{1}{3}(x^3 - 6x^2 + 9x + 9)$$

$$f(x) = x^2 - 4x + 3$$

Globe minimum at $x = 3$

$$\Rightarrow \frac{1}{3}(27 - 54 + 27 + 9) = 3$$

$$\text{Also } f(0) = 3$$

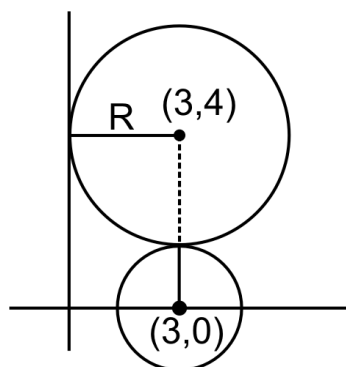
8. Ans (28.00)

$$xy - 3x - 4y + 12 = 0$$

$$(x - 3)(y - 4) = 0 \Rightarrow x = 3 \text{ and } y = 4$$

$$S : x^2 + y^2 - 6x - 8y + c = 0$$

For $R = 3$, $r = 1$ & 7



For $R = 2$, $r = 2$ & 6

For $R = 1$, $r = 3$ & 5

For $R = 0$, $r = 4$